Section XV

NUTRITION & METABOLISM

What is food to one man, may be fierce poison to others.

Lucretius
(99–55 BC)

Chapter 48

Nutritional Requirements

The more impure bodies are fed,. the more diseased they will become

Hippocrates *Aphorisms*

The fundamental goal of nutritional support is to provide the daily nutrient and energy needs of each patient. This chapter describes what is known about providing those needs, according to the most recent clinical practice guidelines (1–3) and expert commentaries (4,5) on the subject. However, it is important to mention at the outset that it is impossible to accurately determine the nutritional needs of critically ill patients, or whether we should be feeding "impure bodies", at all. This latter point (i.e., the notion that feeding critically ill patients can be harmful) is the focus of the very last section of this chapter.

DAILY ENERGY EXPENDITURE

Oxidation of Nutritional Fuels

The energy from the sun is stored in the covalent carbon bonds that hold organic (carbon-based) molecules together. Carbon bonds have differing amounts of stored energy (e.g., double bonds have more energy than single bonds), and molecules that are rich in high-energy carbon bonds are known as *fuels*. The energy in organic fuels is released by a chemical reaction with oxygen, which breaks the carbon bonds and releases the energy. This reaction is known as *oxidation*.

The oxidative metabolism of *nutritional fuels* (i.e., lipids, proteins, and carbohydrates) releases the stored energy and utilizes it to sustain life. This process consumes oxygen, and generates carbon dioxide, water, and heat. The quantities involved in the oxidation of each type of nutritional fuel are shown in Table 48.1. The following points deserve mention.

. The energy yield from nutritional fuels is expressed as heat production (in kilocalories or kcal per gram of fuel).

. The energy yield is highest with lipids (9.1 kcal/g) and lowest with carbohydrates (3.7 kcal/g).

. The respiratory quotient (RQ) is a measure of the CO_2 production (VCO_2) relative to the O_2 consumption (VO_2) (i.e., RQ = VCO_2/VO_2). Carbohydrates have the highest RQ (1.0), and

lipids have the lowest (0.7).

TABLE 48.1 The Oxidation of Nutritional Fuels

Fuel	O_2 Consumption	CO_2 Production	RQ†	Energy Yield
Lipids	2.00 L/g	1.40 L/g	0.7	9.1 kcal/g
Proteins	0.96 L/g	0.78 L/g	0.8	4.0 kcal/g
Carbohydrates	0.74 L/g	0.74 L/g	1.0	3.7 kcal/g

†RQ is "respiratory quotient", and is the ratio of CO_2 production (VCO_2) to O_2 consumption (VO_2): i.e., $RQ = VCO_2/VO_2$.

The oxidative metabolism of all three nutritional fuels determines the whole body VO_2, VCO_2, and RQ. The VO_2 and VCO_2 can be used to determine the energy expenditure (see next) while the whole-body RQ indicates the balance of energy being provided by the nutritional fuels (e.g., an RQ of 1.0 indicates that carbohydrates are the predominant fuel source).

Indirect Calorimetry

It is not possible to directly measure energy expenditure (i.e., heat production) in hospitalized patients. However, it is possible to calculate the resting energy expenditure (REE) from the whole-body VO_2 and VCO_2 using the following equation (6):

$$\text{REE (kcal/min)} = (3.6 \times VO_2) + (1.1 \times VCO_2) - 61 \qquad (48.1)$$

This is the principle of *indirect calorimetry*, which measures the whole-body VO_2 and VCO_2 to derive the energy expenditure.

Methodology

Indirect calorimetry is performed with "metabolic carts" that measure whole-body VO_2 and VCO_2 at the bedside by measuring the concentrations of O_2 and CO_2 in inhaled and exhaled gas. This can be done in spontaneously breathing patients (using a special canopy placed over the patient's head) and during mechanical ventilation (by placing O_2 and CO_2 sensors in line with the ventilator tubing) (7). Steady-state measurements are obtained for 15–30 minutes to determine the REE (kcal/min), which is then multiplied by 1,440 (the number of minutes in 24 hours) to derive the daily energy expenditure (kcal/24 hr). Clinical studies have shown that REE measurements obtained over 30 minutes and extrapolated to 24 hours are equivalent to REE measurements performed for 24 hours (8). The oxygen sensor in the newest indirect calorimetry devices is accurate to inhaled O_2 concentrations up to 70% (9), which is an improvement over the prior upper limit of 60% (10). However, there is still the risk that indirect calorimetry will be unreliable when the inhaled O_2 concentration is above 70%.

All clinical practice guidelines recommend indirect calorimetry for determining the daily energy needs in critically ill patients (1–3). However, indirect calorimetry is not universally

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available, because the equipment is expensive, and it requires trained personnel to operate. When indirect calorimetry is not available, the REE can be estimated, as described next.

A Simple Formula

There are more than 200 cumbersome equations available for estimating daily energy requirements, but none is considered more predictive than the following relationship:

$$\text{REE (kcal/day)} = 25 \times \text{body weight (kg)} \qquad (48.2)$$

This simple predictive relationship is considered suitable for estimating daily energy requirements when indirect calorimetry is not available (2,3). Actual body weight is used unless it is 25% higher than ideal body weight. When actual body weight is more than 125% of ideal body weight, the adjusted body weight is recommended, using the following equation (11):

$$\text{Adjusted weight (kg)} = [(\text{actual} - \text{ideal weight}) \times 0.25] + \text{ideal weight} \qquad (48.3)$$

Recommendations

Some form of nutrition support should be provided within 24–48 hours of ICU admission (1–3). However, the most recent guidelines recommend that nutrition support should begin at REE levels that are less than the measured or estimated REE (1,3). The basis for this recommendation is evidence that endogenous nutritional adjustments (e.g., increased gluconeogenesis) are operative in the early stages of acute illness, and adding a normal nutritional intake to these adjustments results in overfeeding (which can result in hyperglycemia and lipogenesis) (12). In patients who are malnourished at the outset, avoiding aggressive early nutrition will also reduce the risk of the refeeding syndrome (13).

The following recommendations are from the European guidelines for nutrition support (3):

- If indirect calorimetry is used to measure REE, then no more than 70% of the REE should be provided for the first 3 days of nutrition support. Thereafter, the REE can be gradually increased to 100% over the remainder of the first week.
- If predictive equations are used to estimate REE, then no more than 70% of the REE should be provided for the first 7 days of nutrition support.
- In patients who require norepinephrine doses >0.2 μg/kg/min (>14 μg/min in a 70 kg adult), a daily caloric intake of 10–15 kcal/kg/day should be considered for 7 days, or until the clinical condition improves (5).

SUBSTRATE REQUIREMENTS

The daily energy requirement should be provided by nonprotein calories from carbohydrates and lipids, while protein intake is used to maintain essential enzymatic and structural proteins.

Carbohydrates

Standard nutrition regimens use carbohydrates to provide about 70% of the nonprotein calories. The human body has limited carbohydrate stores (Table 48.2), and daily intake of carbohydrates

is considered necessary to ensure proper functioning of the central nervous system, which uses about 100–120 grams of glucose daily (3). Therefore, a daily carbohydrate intake of at least 130 grams is advised (3).

Physiological stress is common in critically ill patients, and this promotes insulin resistance and hyperglycemia (14). In addition, excessive carbohydrate intake promotes CO_2 production (see Table 48.1) and lipogenesis (14). For these reasons, carbohydrate intake should be limited to no more than 5 mg/kg/min (3), which is about 500 grams daily in a 70 kg adult.

TABLE 48.2 Endogenous Fuel Stores in Healthy Adults

Fuel Source	Amount (kg)	Energy Yield (kcal)
Adipose Tissue Fat	15.0	141,000
Muscle Protein	6.0	24,000
Total Glycogen	0.09	900 ------ Total: 165,000

Data from Cahill GF. Jr. N Eng J Med 1970; 282:668–675.

Lipids

Standard nutrition regimens use lipids to provide approximately 30% of the daily energy needs. Dietary lipids have the highest energy yield of the three nutrient fuels (Table 48.1), and lipid stores in adipose tissues represent the major endogenous fuel source in healthy adults (Table 48.2).

Linoleic Acid

Dietary lipids are triglycerides, which are composed of a glycerol molecule linked to three fatty acids. The only dietary fatty acid that is considered essential (i.e., must be provided in the diet) is linoleic acid, a long chain, polyunsaturated fatty acid (PUFA). A deficient intake of this essential fatty acid produces a clinical disorder characterized by a scaly dermopathy, cardiac dysfunction, and increased susceptibility to infections (15). This disorder is prevented by providing 0.5% of the dietary fatty acids as linoleic acid. Soybean oil is used as the source of linoleic acid in most nutritional support regimens.

Propofol

Propofol, an intravenous anesthetic agent that is popular for short-term sedation in the ICU, is mixed in a 10% lipid emulsion that provides 1.1 kcal/mL (2,3). As a result, *the calories provided by propofol infusions must be considered when calculating the nonprotein calories in a nutrition support regimen*. Propofol infusions can also promote hypertriglyceridemia (16), so triglyceride levels should be monitored during prolonged infusions of propofol.

Excessive Lipid Intake

Excessive intake of lipids, especially PUFAs (which are highly oxidizable), can lead to oxidative damage in tissues, especially the lungs (17), and can impair immune function (3). To prevent

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lipid overload, *the recommended maximum lipid intake is 1.5 grams/kg per day* (3). (*Note:* It is also possible to adjust the type of lipid intake to reduce the risk of inflammatory injury. This is described in chapters 49 and 50.)

Protein Requirements

The normal protein requirement is 0.8–1.0 grams/kg/day in healthy patients, but because hypercatabolism is common *in critically ill patients, a protein intake of 1.2– 2.0 grams/kg/day is recommended* (1,2).

Nitrogen Balance

The adequacy of protein intake can be evaluated with the nitrogen balance; i.e., the difference between intake and excretion of protein-derived nitrogen. However, *patients should be in a stable condition before this evaluation is performed.*

. *Nitrogen Excretion:* Two thirds of the nitrogen derived from protein breakdown is excreted in the urine (18), and about 85% of this nitrogen is contained in urea (the remainder is in ammonia and creatinine). Thus, the urinary urea nitrogen (UUN), measured in grams excreted in 24 hours, represents the bulk of nitrogen derived from protein breakdown. The remainder of the protein-derived nitrogen (usually about 4–6 grams/day) is excreted in the stool. Therefore, protein-derived nitrogen excretion can be expressed as follows:

$$\text{Nitrogen Excretion (g/24 h)} = \text{UUN} + (4\text{–}6) \qquad (48.4)$$

In the presence of diarrhea, non-urinary nitrogen losses cannot be estimated accurately, and nitrogen balance determinations are unreliable.

. *Nitrogen Intake:* Protein is 16% nitrogen, so each gram of protein contains 1/6.25 grams of nitrogen. Therefore, protein-derived nitrogen intake is derived as follows:

$$\text{Nitrogen Intake (g/24 h)} = \text{Protein Intake (g/24 h)}/6.25 \qquad (48.5)$$

. *Nitrogen Balance:* The equations for nitrogen intake and nitrogen excretion are combined to determine the daily nitrogen balance.

$$\text{Nitrogen Balance (g/24 h)} = \text{Protein Intake}/6.25 - [\text{UUN} + (4\text{–}6)] \qquad (48.6)$$

. The goal of nutritional support is a positive nitrogen balance of 4–6 grams.

Nitrogen Balance & Nonprotein Calories

The first step in achieving a positive nitrogen balance is to provide enough nonprotein calories to spare proteins from being degraded to provide energy. This is demonstrated in Figure 48.1. When the daily protein intake is constant, the nitrogen balance becomes positive only when the intake of nonprotein calories is sufficient to meet the daily energy needs (i.e., the REE). Therefore, *increasing protein intake will not achieve a positive nitrogen balance unless the intake of nonprotein calories is adequate*.

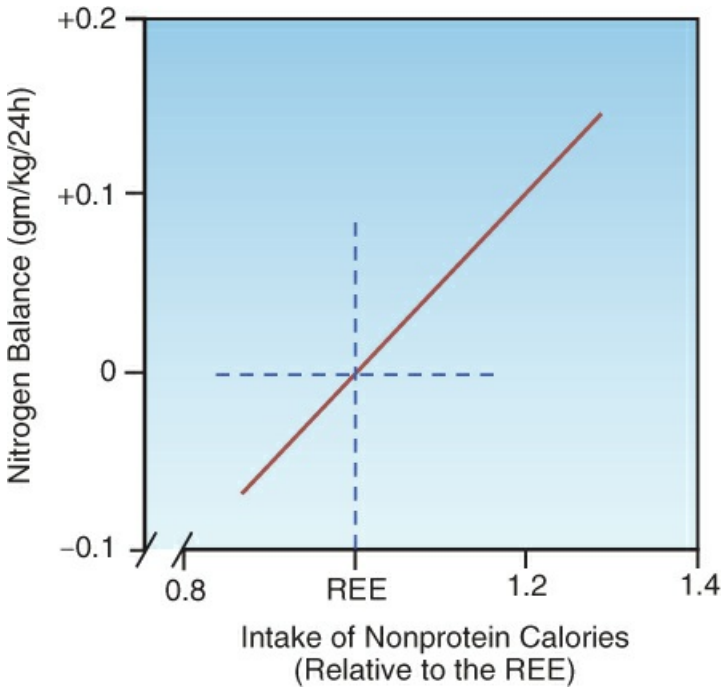


FIGURE 48.1 Relationship between nitrogen balance and the daily intake of nonprotein calories relative to daily calorie requirements. Protein intake is constant. REE = resting energy expenditure.

VITAMIN REQUIREMENTS

Thirteen vitamins are considered an essential part of the daily diet, and Table 48.3 shows the recommended daily requirement for these vitamins in healthy adults. Daily vitamin requirements are likely to be higher in critically ill patients, and will also vary in individual patients. The following is a description of vitamin deficiencies that deserve particular attention in critically ill patients.

TABLE 48.3 Daily Vitamin Requirements in Adults, Ages 51–70

Vitamin	Men	Women*
Vitamin A	900 µg/d	700 µg/d
Vitamin C	90 mg/d	75 mg/d
Vitamin D	600 IU/d	600 IU/d
Vitamin E	15 mg/d	15 mg/d
Vitamin K	120 µg/d	90 µg/d
Thiamine	1.2 mg/d	1.1 mg/d
Riboflavin	1.3 mg/d	1.1 mg/d
Niacin	16 mg/d	14 mg/d

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Pyridoxine	1.7 mg/d	1.5 mg/d
Folate	400 µg/d	400 µg/d
Vitamin B_{12}	2.4 µg/d	2.4 µg/d
Pantothenic Acid	5 mg/d	5 mg/d
Biotin	30 µg/d	30 µg/d

From the National Institutes of Health, Office of Dietary Supplements. Available at ods.od.nih.gov/HealthInformation/nutrientrecommendations.aspx. Accessed 7/2/2024 *Does not include pregnant or lactating women.

Thiamine Deficiency

Thiamine (vitamin B_1) plays an essential role in carbohydrate metabolism, because thiamine is converted to thiamine pyrophosphate, which is a cofactor for pyruvate dehydrogenase, the enzyme that allows pyruvate to enter mitochondria and generate high-energy ATP molecules (see Figure 9.5) (19). Thiamine deficiency thus creates an energy deficit from impaired glucose metabolism. (There are other metabolic consequences of thiamine deficiency, which are beyond the scope of this text.) Healthy adults have a limited store of thiamine (about 30 mg), which is depleted after 2–3 weeks in the absence of thiamine intake (19). The normal thiamine requirement is about 1 mg/day in healthy adults (see Table 48.3).

Predisposing Factors

There are several conditions in ICU patients that promote thiamine deficiency, and the major ones are listed in Table 48.4. Alcoholism is considered the major cause of thiamine deficiency; an effect that is attributed to alcohol's ability to inhibit thiamine absorption from the GI tract (20). Thiamine deficiency has also been reported in as many as 70% of ICU patients with septic shock (21). Other predisposing factors that are common in ICU patients include the chronic use of furosemide (22), which promotes thiamine excretion by the kidneys, and magnesium depletion (23), which inhibits the conversion of thiamine to thiamine pyrophosphate (24). Thiamine deficiency has also been reported following gastric surgery for obesity (25), presumably as a result of impaired thiamine absorption from the GI tract.

REFEEDING: Patients who are admitted with borderline thiamine stores can become thiamine deficient when they receive a glucose load (usually from nutrition support regimens). This is similar in principle to the refeeding syndrome, although it is a distinct entity (26).

TABLE 48.4 Thiamine Deficiency

Predisposing Factors	Clinical Manifestations
• Alcohol Abuse • Malnutrition • Septic Shock • Furosemide • Mg^{++} Depletion	• Cardiomyopathy (Beriberi) • Encephalopathy (Wernicke's) • Peripheral Neuropathy • Lactic Acidosis

Treatment[†]

1. 500 mg thiamine IV every 8 hours for 2–3 days, then
2. 250 mg thiamine IV once daily for 3–5 days, then
3. 100 mg thiamine PO three times daily for 1–2 weeks, then
4. 100 mg thiamine PO once daily for maintenance therapy.

Clinical Manifestations

The clinical manifestations of thiamine deficiency are also common occurrences in ICU patients (see Table 48.4), and include: (*a*) a cardiomyopathy, also known as "wet beriberi", (*b*) Wernicke's encephalopathy, (*c*) a peripheral neuropathy, also known as "dry beriberi", and (*d*) lactic acidosis.

CARDIOMYOPATHY: The cardiomyopathy associated with thiamine deficiency may be the result of increased oxidative stress (27), and it typically produces a high output heart failure (28). Thiamine deficiency from chronic furosemide use could also aggravate other forms of cardiomyopathy, although thiamine supplementation in patients with heart failure has no convincing effects on cardiac function or symptomatology (29).

ENCEPHALOPATHY: Wernicke's encephalopathy is the most cited consequence of thiamine deficiency. The clinical presentation is classically described as the combination of mental status changes, ocular abnormalities, and ataxia, but this "triad" occurs in only 20% of cases (30). About 80% of patients have mental status changes (e.g., somnolence, confusion, stupor), and 30% have ocular abnormalities (e.g., nystagmus, lateral gaze paralysis) (30). Magnetic resonance imaging (MRI) can show characteristic abnormalities, like the ones in Figure 48.2, and these may be the only clue to the diagnosis (see later).

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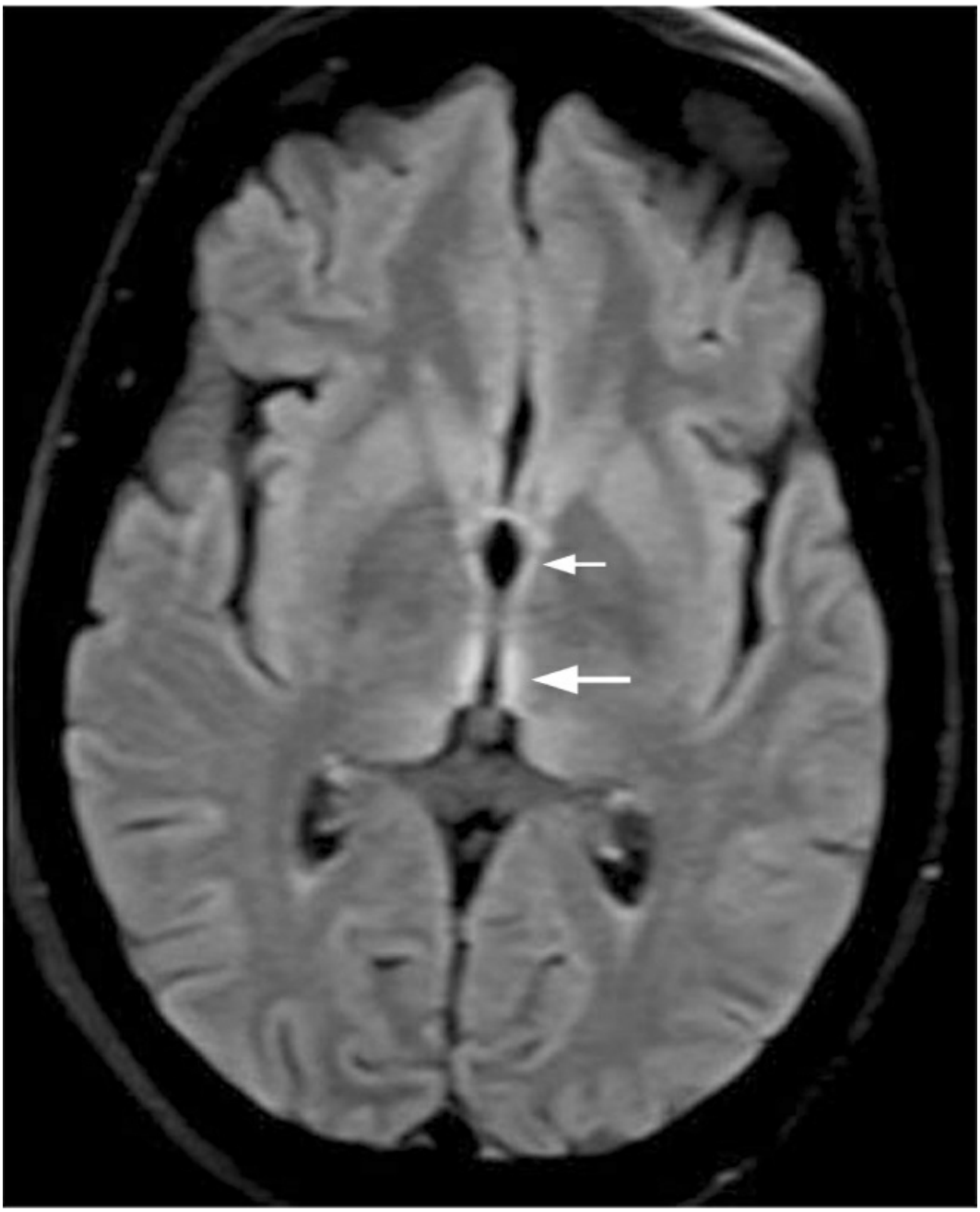


FIGURE 48.2 FLAIR imaging on MRI showing hyperdense signals bilaterally in the medial thalamus (large arrow) and the periventricular region of the third ventricle (small arrow) in a patient with Wernicke's encephalopathy. Case courtesy of Ashutosh Gandhi, Radiopaedia.org, rID: 19865.

LACTIC ACIDOSIS: As explained, thiamine deficiency impairs the movement of pyruvate into mitochondria, and this results in an increase in lactate formation (see Figure 9.5). Thiamine deficiency may be responsible for some of the lactic acidosis associated with sepsis (31), and for the lactate elevation in diabetic ketoacidosis (32). Lactate elevation of unclear etiology should always prompt suspicion of thiamine deficiency.

PERIPHERAL NEUROPATHY: Relatively little is written about the peripheral neuropathy in thiamine deficiency, other than it often coexists with Wernicke's encephalopathy (30).

Diagnosis

The diagnosis of thiamine deficiency is challenging. Plasma thiamine levels are unreliable. Most of the thiamine in the body is present as thiamine pyrophosphate (TPP), which is located predominantly in red blood cells, so a whole blood assay for TPP has some merit for evaluating thiamine status. The normal level is 63–229 nmol/L (30). However, this test is not available in many clinical laboratories, and results are often not immediately available.

Because of the limitations in the laboratory evaluation of thiamine status, thiamine deficiency is considered a clinical diagnosis (30). Unfortunately, the manifestations of thiamine deficiency are common occurrences in critically ill patients. Thiamine deficiency should be suspected when one of the manifestations (heart failure, encephalopathy, or lactic acidosis) has no apparent etiology, especially in the presence of a predisposing condition (e.g., alcohol abuse).

IMAGING: MRI is considered the most reliable method for confirming the diagnosis of Wernicke's encephalopathy (30). Common abnormalities include an increased T2 signal bilaterally in the medial regions of the thalamus (see Figure 48.2), and in the hypothalamus, mamillary bodies, and midline regions of the cerebellum (30). MRI has a sensitivity of 53% and a specificity of 93% for the diagnosis of Wernicke's encephalopathy (30).

Treatment

Patients who are considered at risk for thiamine deficiency (e.g., alcoholics) are often given *thiamine in a conventional dose of 100 mg daily* (33), but this *is inadequate for correcting thiamine deficiency* (33). The recommended treatment for suspected or confirmed cases of thiamine deficiency is shown in Table 48.4 (33,34). The intravenous route is always preferred for acute thiamine replacement, as thiamine uptake from the GI tract is often erratic in critically ill patients. When replenishing thiamine, it is important to *identify and correct magnesium deficiency*, so the infused thiamine can be utilized (by conversion to thiamine pyrophosphate).

Vitamin D Deficiency

Vitamin D deficiency is reported in 40–100% of ICU patients (35). This is not surprising because the recommended daily requirement for vitamin D is 600 IU for adults (and is likely to be even higher in critically ill patients), yet enteral and parenteral nutrition regimens typically provide only about 200–400 IU of vitamin D daily.

Consequences

The major consequence of vitamin D deficiency is bone health, which should have a relatively minor impact in critically ill patients. However, vitamin D is also involved in regulation of the immune response, and vitamin D deficiency is associated with increased rates of infection in ICU patients (36).

Diagnosis

The diagnosis of vitamin D deficiency is based on the plasma concentration of a vitamin D

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metabolite, 25-hydroxyvitamin D (25-OHD). Plasma levels below 50 nmol/L indicate a deficiency state, and plasma levels below 30 nmol/L indicate a severe deficiency (37).

TREATMENT: The reliability of oral vitamin D supplementation in critically ill patients is unclear. A single IM injection of 150,000 IU of cholecalciferol can increase plasma 25-OHD levels to normal in 90% of ICU patients (37).

Antioxidant Vitamins

The following vitamins are considered major endogenous antioxidants that protect tissues against oxidative injury (38).

. *Vitamin E* (α-tocopherol) is the major lipid-soluble antioxidant in the body, and protects all cell membranes from oxidative damage (39).

. *Vitamin C* (ascorbic acid) is a major water-soluble antioxidant, and scavenges (detoxifies) reactive oxygen species. It is also essential for maintaining vitamin E in its active form (40).

Both antioxidant vitamins can be deficient in critically ill patients (41,42) and there is evidence that supplementing vitamin C and vitamin E can improve outcomes (43), although this is not a consistent finding (38). Nevertheless, considering that oxidative stress plays an important role in the pathogenesis of inflammatory-mediated organ injury (44), it seems wise to maintain adequate antioxidant protection in critically ill patients.

ESSENTIAL TRACE ELEMENTS

A trace element is one that is present in amounts less than 50 μg per gram of body tissue (45). Seven trace elements are considered essential in humans (i.e., are associated with deficiency syndromes), and these are listed in Table 48.5 along with the daily requirements for each element in healthy men and women. As mentioned for vitamins, the essential trace element requirements are likely to be higher in critically ill patients. The following trace elements deserve mention because of their relevance to oxidative tissue injury.

Iron

Healthy adults have about 4.5 grams of iron, but very little of this is free, unbound iron (46). Most of the iron in the body is bound to hemoglobin, and the remainder is bound to ferritin in tissues and transferrin in plasma. Circulating transferrin is only 30% saturated with iron, so any increase in iron will be quickly bound by transferrin, thus preventing a rise in free iron in plasma. The paucity of free iron in the body has survival value, because free iron has a pivotal role in oxidative tissue injury (47).

Iron and Oxidative Injury

Iron has two actions that promote oxidative cell injury.

. Iron in its reduced form (Fe^{2+}) is necessary for the production of hydroxyl radicals (see Figure 25.6), which are the most destructive of the reactive oxygen species, capable of damaging all

vital cell components.

. Iron-induced lipid peroxidation can produce a form of regulated cell death known as *ferroptosis* (48). This process is triggered by inflammation, and it is has a major role in inflammatory tissue injury (49).

The risk of harm from free iron would explain why hypoferremia is common in conditions associated with systemic inflammation (50). This risk is also reason to avoid administering iron in critically ill patients unless there is evidence of iron depletion in tissues with resultant anemia. Tissue iron stores can be assessed with the plasma ferritin level; i.e., a ferritin level below 18 µg/L indicates probable iron deficiency, while iron deficiency is unlikely if the ferritin level exceeds 100 µg/L (51).

TABLE 48.5 Daily Requirements for Essential Trace Elements in Adults, Ages 51–70

Element	Men	Women†
Chromium	30 µg/d	20 µg/d
Copper	900 µg/d	900 µg/d
Iodine	150 µg/d	150 µg/d
Iron	8 mg/d	8 mg/d
Manganese	2.3 mg/d	1.8 mg/d
Selenium	55 µg/d	55 µg/d
Zinc	11 mg/d	8 mg/d

From the National Institutes of Health, Office of Dietary Supplements. Available at ods.od.nih.gov/HealthInformation/nutrientrecommendations.aspx. Accessed 7/2/2024. †Does not include pregnant or lactating women.

Selenium

Selenium is an essential cofactor for the enzyme glutathione peroxidase, which is a major source of intracellular antioxidant activity (see Figure 25.7). The daily requirement for selenium is 55 µg in healthy adults (see Table 48.5), but selenium utilization is increased in acute illness (52), and selenium deficiency is common in critically ill patients (45).

Plasma selenium levels can be monitored in patients at risk for oxidative stress (e.g., from systemic inflammation). Plasma levels below 100 µg/L are associated with reduced activity of glutathione peroxidase (53). Selenium supplementation can be given intravenously (as sodium selenite), and the maximum dose considered safe is 225 µg daily (54).

A FINAL WORD

A Fly in the Ointment

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The rationale for providing nutrition support in critically ill patients is to prevent or alleviate malnutrition. This is a valid reason if the malnutrition is due to decreased nutrient intake (i.e., starvation), but *the malnutrition associated with critical illnesses is the result of abnormal nutrient processing*, and providing nutrients will not correct this type of malnutrition until the causative illness is corrected, and the metabolic derangements are resolved. In fact, nutrition support can be harmful in this situation (55), as explained next.

An example of abnormal nutrient processing in acute illness is the fate of a glucose load; i.e., less than 5% of glucose is metabolized to lactate in healthy subjects, while as much as 85% of a glucose load can be recovered as lactate in acutely ill patients (56). This is demonstrated in Figure 48.3 (57). In this case, patients undergoing a major surgical procedure were randomized to receive either Ringer's lactate or a 5% dextrose solution as the IV fluid during the procedure. As indicated in the graph, the dextrose infusions (which averaged 200 grams) resulted in an increase in the plasma lactate to abnormally high levels (>2 mmol/L), which did not occur in the patients who received a dextrose-free fluid.

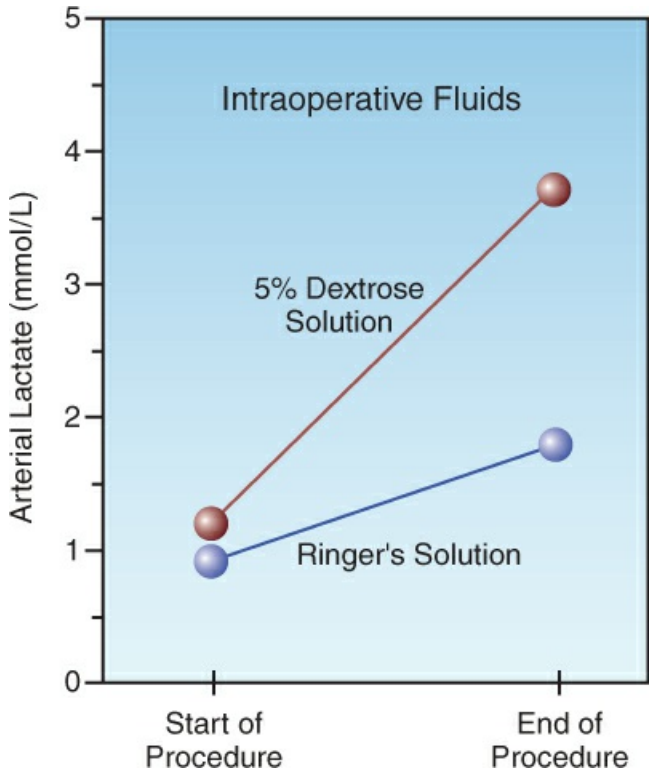


FIGURE 48.3 Influence of dextrose infusions on arterial lactate levels during abdominal aortic surgery. Each data point represents a mean value for patients randomized to receive either Ringer's lactate or a 5% dextrose solution during the procedure. Total volume infused was equivalent with both fluids. Data from Reference 57.

Another potential for "nutrient harm" would be the infusion of lipids in conditions that favor oxidation (e.g. inflammation), because oxidation of the infused lipids can be a source of tissue injury (17), as mentioned earlier.

To summarize, because of the metabolic derangements in critical illness, nutrition support is unlikely to provide the desired nutritional benefit, and can instead be a source of harm. Maybe

this is why we lose our appetite when we get sick?

It seems Hippocrates had the right idea about 2,500 years ago.

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Thiamine Deficiency

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A L G r a w a n y

Chapter 49

Enteral Nutrition

Forced feeding . . . goes on because the world still believes it can eat itself well.

Herbert Shelton (*a*)

For patients who are unable to eat, the preferred method of nutrition support is the infusion of liquid feeding formulas into the stomach or small bowel. This not only provides energy from nutrient fuels, it also serves as an *infection control measure* by promoting the barrier function of the bowel mucosa (as explained in the next section).

This chapter presents the practical aspects of providing enteral tube feedings: i.e., when to do it and when not to do it, how to do it, what feeding formulas to use, and what complications to watch for. When possible, recommendations are taken from clinical practice guidelines and recent reviews of the topic (1–5).

GENERAL CONSIDERATIONS

Infectious Risk

The threat of sepsis and organ injury from the bowel is described in Chapter 4. The primacy of this threat is evident in the description of the bowel as "the motor of multiorgan failure" (6). The role of enteral tube feedings in mitigating this threat is summarized in the following statements:

. The presence of bulk nutrients in the lumen of the bowel has a trophic effect on the bowel mucosa that preserves the structural integrity of the mucosa (7). This maintains the barrier function of the bowel mucosa, which protects against invasion from enteric pathogens via *translocation* across the bowel wall (8).

. The trophic influence of enteral nutrition also includes immune defenses in the bowel, such as the production of immunoglobulin A (IgA) by monocytes in the bowel wall, which blocks the attachment of pathogens to the bowel mucosa (9).

. The trophic effect of enteral nutrition requires nutritional bulk rather than the presence of nutrient fuels (i.e., carbohydrates, proteins and lipids), because the oral administration of

parenteral feeding solutions does not produce the effect (10). However, glutamine is believed to be involved in trophism because it is the principal fuel for enterocytes in the bowel mucosa (11).

. The trophic effect of nutritional bulk is lost during periods of bowel rest, which is accompanied by progressive atrophy of the bowel mucosa (6), along with an increased risk of bacterial invasion from the bowel (12).

The sum of these observations indicates that the normal antimicrobial defenses in the bowel are sustained by enteral tube feedings, but not by parenteral nutrition regimens. In this sense, the delivery of enteral tube feedings can be viewed as an infection control measure. This is verified by clinical studies showing *a lower infection rate associated with early institution of enteral tube feedings* (i.e., within 48 hours of admission) (4).

Who and When

The enteral route is always preferred to the intravenous route for nutrition support, and when oral food intake is not possible, *enteral tube feedings should be started within 48 hours of admission* (1–4), unless there is a contraindication to enteral nutrition (see next). *The presence of bowel sounds is not required to initiate enteral tube feedings* (13).

Contraindications

The contraindications to enteral tube feedings are listed in Table 49.1. One concern that receives the most attention is the wisdom of enteral tube feedings in patients with circulatory shock who require a vasoconstrictor drug (a vasopressor) for blood pressure support. The current consensus about this concern is summarized below.

. Enteral feeding is not advised in uncontrolled or progressive circulatory shock.

. Stable or decreasing vasopressor infusions are not a contraindication to enteral nutrition. However, if the infusion rate of norepinephrine is >0.2 µg/kg/min (>14 µg/min in a 70 kg adult), a reduced caloric intake of 10–15 kcal/kg/day (instead of the usual 25 kcal/kg/day) is recommended for the first 7 days, or until the clinical condition improves (14).

TABLE 49.1 Contraindications to Enteral Tube Feedings

1. Uncontrolled circulatory shock
2. Bowel ischemia (suspected or proven)
3. Life-threatening, refractory hypoxemia or hypercapnia
4. Active upper GI hemorrhage
5. Bowel obstruction
6. High-output intestinal fistula
7. Abdominal compartment syndrome
8. Gastric residual volume >500 mL in 6 hours

From Reference 3.

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FEEDING FORMULAS

There is a multitude of tube feeding formulas (Abbott Laboratories, the leading source of enteral nutrition formulas in the United States, has 92 different products) (15), but only a small number of formulas are used routinely. The following is a brief description of some characteristic features of feeding formulas. Representative formulas are included in Tables 49.2 and 49.3.

Caloric Density

Feeding formulas are available with caloric densities of 1, 1.2, 1.5, and 2 kcal/mL. Most tube feeding regimens use formulas with a low caloric density (1 or 1.2 kcal/mL), because higher caloric densities have higher osmolalities, which promotes diarrhea. The higher caloric density formulas (1.5 and 2 kcal/mL) are used when fluid restriction is a priority.

TABLE 49.2 Representative Tube Feeding Formulas

Feeding Formula	Caloric Density (kcal/mL)	Protein (g/L)	Fiber (g/L)	Osmolality (mosm/kg H_2O)
Isocal	1.0	35	0	324
Vital 1.0[5]	1.0	40	4	411
Osmolite 1.2	1.2	56	0	360
Osmolite 1.5	1.5	63	0	525
Jevity 1.2	1.2	56	17	450
Twocal HN	2.0	84	5	710

[5]An elemental formula, with hydrolyzed proteins and medium chain triglycerides to facilitate absorption.
All formulas, except Isocal are from the Abbott Nutrition Product Guide (15).

Protein Content

The protein content in standard feeding formulas varies from 35 to 65 g/L. (The daily protein requirement for an ICU patient is about 1.5 g/kg, which is 100 grams/day for a 70 kg adult.) Protein-enriched formulas are often designated by the suffix HN (for “high nitrogen”), and they provide about 20% more protein than the standard formulas (see Twocal HN in Table 49.2).

Most enteral formulas contain intact proteins that are broken down into amino acids in the upper GI tract. These are called *polymeric* formulas. There are also feeding formulas where the protein is already broken down into small peptides (called *semi-elemental* formulas) or individual amino acids (called *elemental* formulas) to facilitate absorption from the bowel. This

also promotes water reabsorption from the bowel, so these formulas can be advantageous in patients with troublesome diarrhea. *Vital 1.0* in Table 49.2 is an elemental formula.

Carbohydrate Content

Carbohydrates (usually polysaccharides) are the major source of calories in feeding formulas, and they provide up to 70% of the total calories.

Fiber

The term "fiber" refers to polysaccharides from plants that are not digested by humans. There are two types of fiber: fermentable, and nonfermentable.

. *Fermentable fiber* is broken down by gut bacteria to produce short-chain fatty acids (butyrate being the most important) that are the principal fuel source for the cells of the colonic mucosa (16). These fatty acids also promote the growth of nonpathogenic bowel microbes (17), which helps to prevent colonization with pathogens. Thus, fermentable fiber promotes the "health" of both the microbiome, and the mucosal barrier in the large bowel.

. *Nonfermentable fiber* is not broken down by gut bacteria, and the undigestible residue creates an osmotic force that draws water into the lumen of the bowel. This increases the water content of stool to facilitate laxation.

The fiber in liquid feeding formulas is a mixture of fermentable and nonfermentable fiber. The feeding formula with the highest fiber content is *Jevity* (see Table 49.2).

Lipid Content

Standard feeding formulas contain polyunsaturated fatty acids (PUFAs) from vegetable oils. The lipid content of feeding formulas typically provides about 30% of the total calories.

Omega-3 Fatty Acids

PUFAs from vegetable oils can serve as precursors for inflammatory mediators (eicosanoids) that are capable of promoting inflammatory cell injury. This concern has prompted the introduction of feeding formulas that contain PUFAs from fish oils (omega-3 fatty acids), which do not promote inflammation. Some of these feeding formulas are shown in Table 49.3. The use of feeding formulas that influence the inflammatory response is known as *immunonutrition* (14).

TABLE 49.3 Immune-Modulating Feeding Formulas

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Feeding Formula	Calories (kcal/mL)	ω-3 FAs (g/L)	Arginine (g/L)	Antioxidants
Impact	1.0	1.7	13	Selenium, β-carotene
Optimental	1.0	2.3	6	Vitamins C & E, β-carotene
Oxepa	1.5	4.6	0	Vitamins C & E, β-carotene

ω-3 FAs = omega-3 fatty acids

Fish oil-enriched feeding formulas should not be used routinely (3), but there is some evidence of benefit when they are used in patients with acute respiratory distress syndrome (ARDS) (18). Other conditions that might benefit from immunonutrition are major trauma, and major abdominal surgery (19).

Conditionally Essential Nutrients

Non-essential nutrients can become essential (i.e., require exogenous support) in conditions of increased utilization. Two of these *conditionally essential* nutrients deserve mention.

Arginine

Arginine is a preferred metabolic substrate for injured muscle, and it is also a precursor of nitric oxide (which has a pivotal role in maintenance of peripheral blood flow) (20). Arginine can become depleted in conditions like multisystem trauma, due to increased utilization in skeletal muscle (20). At least 8 enteral feeding formulas contain arginine in concentrations of 8–19 g/L (two of these formulas are included in Table 49.3), although the optimal dose of arginine is not known. Arginine-enriched feedings have a documented benefit (i.e., reduced risk of infection) in the perioperative period for major surgery (21).

RECOMMENDATION: Clinical practice guidelines state that arginine-enriched feedings are a "consideration" in patients with traumatic brain injury, and in perioperative patients in the surgical ICU (2). They should not be used in medical ICU patients, especially those with severe sepsis (because of the risk of hypotension from arginine-initiated production of nitric oxide).

Glutamine

Glutamine is the most abundant free amino acid in the body, and has several functions that have survival value (22). It is a precursor for glutathione, the major intracellular antioxidant, and is essential for nucleic acid synthesis, and for the production of select neurotransmitters (e.g., gamma aminobutyric acid). Glutamine is also an oxidative fuel for cells with a rapid turnover, like the bowel mucosa. Glutamine stores are rapidly depleted in conditions that promote protein catabolism (e.g., sepsis, major trauma, burn injury), and glutamine depletion has a negative impact on outcomes (23).

Glutamine is contained in the proteins provided by tube feeding formulas, but additional

glutamine is recommended in the following conditions (3)

. For critically ill trauma patients, additional glutamine doses of 0.2–0.3 g/kg/day should be added to the tube feedings for the first 5 days of enteral nutrition.
. For patients with burns over more that 20% of the body surface area, additional glutamine doses of 0.3–0.5 g/kg/day should be added to the tube feedings for the first 10–15 days of enteral nutrition.

Glutamine supplementation is not recommended in other conditions.

Carnitine

Carnitine is necessary for the transport of long-chain fatty acids into mitochondria for fatty acid oxidation (24). Carnitine deficiency can be the result of a genetic disorder, malnutrition, hypercatabolic states, and liver failure, and manifestations include hypoglycemia, and a cardiomyopathy (25). A deficiency state is suggested by plasma carnitine levels below 20 mmol/L.

The recommended daily dose of carnitine is 20–30 mg/kg in adults (26). Feeding formulas that provide supplemental carnitine include *Glucerna, Isocal HN, Jevity, and Peptamen*.

Overabundance of Feeding Formulas

Despite the multitude of feeding formulas, there is little evidence that one formula, or one type of formula, produces better outcomes than the others. As a result, there is a limited number of formulas available in each hospital, and you will probably use the same 1 or 2 formulas for most of your patients.

CREATING A FEEDING REGIMEN

This section describes a simple method for creating an enteral feeding regimen. This method is summarized in Table 49.4, and proceeds in four steps.

TABLE 49.4 Creating an Enteral Feeding Regimen

A L G r a w a n y

Step 1a. Determine the resting energy expenditure (REE).

Measure the REE with indirect calorimetry (recommended), or estimate the REE as follows:.

$$\text{REE (kcal/day)} = 25 \times \text{wt (kg)}$$

Step 1b. Estimate the daily protein requirement:

$$\text{Protein (g/day)} = (1.2\text{–}1.6) \times \text{wt (kg)}$$

Step 2. Select the feeding formula.

If possible, use a feeding formula with a caloric density of 1–1.2 kcal/mL.

Step 3. Determine the infusion rate:

$$\text{a. Feeding volume (mL)} = \frac{\text{kcal/day required}}{\text{kcal/mL in feeding formula}}$$

For continuous infusions:

$$\text{b. Infusion rate (mL/hr)} = \frac{\text{Feeding volume (ml)}}{\text{24 (hr)}}$$

Step 4. Adjust the protein intake, if necessary:

a. Determine the projected daily protein intake:

$$\text{Protein In (g/day)} = \text{Feeding volume (L/day)} \times \text{protein (g/L) in formula}$$

b. If the projected intake is less than desired intake, add powdered protein to the feeding solution to correct the discrepancy

Step 1. Determine the daily energy and protein requirements.

The very first consideration is the patient's daily requirement for calories and protein. The daily calorie requirement, also called the *resting energy expenditure* (REE) should be measured using *indirect calorimetry*, which is described in Chapter 48. However, if this is not available, the REE can be estimated using the following simple relationship:

$$\text{REE (kcal/d)} = 25 \times \text{body weight (kg)} \qquad (49.1)$$

The actual body weight is used for this determination, although the dry body weight seems more appropriate. If actual body weight exceeds 125% of ideal body weight, then the "adjusted body weight" can be used for the determination (as shown in Equation [48.3]).

The recommended daily protein requirements in critically ill patients is 1.2 to 2.0 grams/kg/day (1,2). (A median value of 1.6 g/kg/day is appropriate.) When the condition stabilizes, a nitrogen balance can be performed to determine if the protein intake is adequate (as described in Chapter 48). However, this is rarely done while patients are in the ICU.

Step 2. Select a feeding formula.

A standard formula, with 1–1.2 kcal/mL, should be sufficient for most patients. Use formulas with higher caloric densities only if volume restriction is a priority.

Step 3. Calculate the desired infusion rate.

A continuous infusion is recommended for enteral tube feeding (instead of bolus feedings). To determine the desired infusion rate, first calculate the volume of the feeding formula that must be infused to meet the daily caloric requirement, as shown in Table 49.4. The infusion rate (mL/hr) is then the volume to be infused over 24 hours, divided by 24.

Propofol Adjustment

Propofol is a popular agent for short-term sedation during mechanical ventilation, and is delivered in a lipid emulsion that has a caloric density of 1.1 kcal/mL. The calories provided by propofol infusions should be subtracted from the estimated or measured REE to determine the calories that must be provided by the diet. Since propofol provides 1 kcal/mL, the hourly infusion rate of propofol (mL/hr) is equivalent to the hourly caloric yield (kcal/hr).

Step 4. Adjust the protein intake, if necessary.

The final step in the process is determining if the feeding regimen you just created will satisfy the daily protein requirement (from Step 1b). The projected protein intake is simply the daily feeding volume multiplied by the protein concentration in the feeding formula. If the projected protein intake is less than required, powdered protein can be added to the tube feedings to correct the discrepancy.

The Initial Period

The most recent guidelines recommend that *nutrition support should begin at REE levels that are less than the measured or estimated REE* (1,3). This recommendation is based on the discovery that endogenous nutritional adjustments (e.g., gluconeogenesis) are operative in the early stages of acute illness, and adding full nutrition support to these adjustments results in overfeeding (27). In patients who are malnourished at the outset, aggressive early nutrition also increases the risk of the refeeding syndrome (28).

The following recommendations are from the European guidelines for nutrition support (3):

. If indirect calorimetry is used to measure REE, then no more than 70% of the REE should be

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provided for the first 3 days of nutrition support. Thereafter, the REE can be gradually increased to 100% over the remainder of the first week.

. If predictive equations are used to estimate REE, then no more than 70% of the REE should be provided for the first 7 days of nutrition support.

FEEDING TUBES

Liquid feeding formulas are delivered into the stomach or duodenum using narrow-bore, flexible nasogastric (NG) tubes like the one shown in Figure 49.1. These tubes are typically 8–10 French in diameter, and 20–40 inches (50–100 cm) in length. They are made of polyurethane, and are much more flexible than the polyvinylchloride NG tubes that are used to decompress the stomach. Because of their flexibility, a guidewire or stylet is necessary for placement.

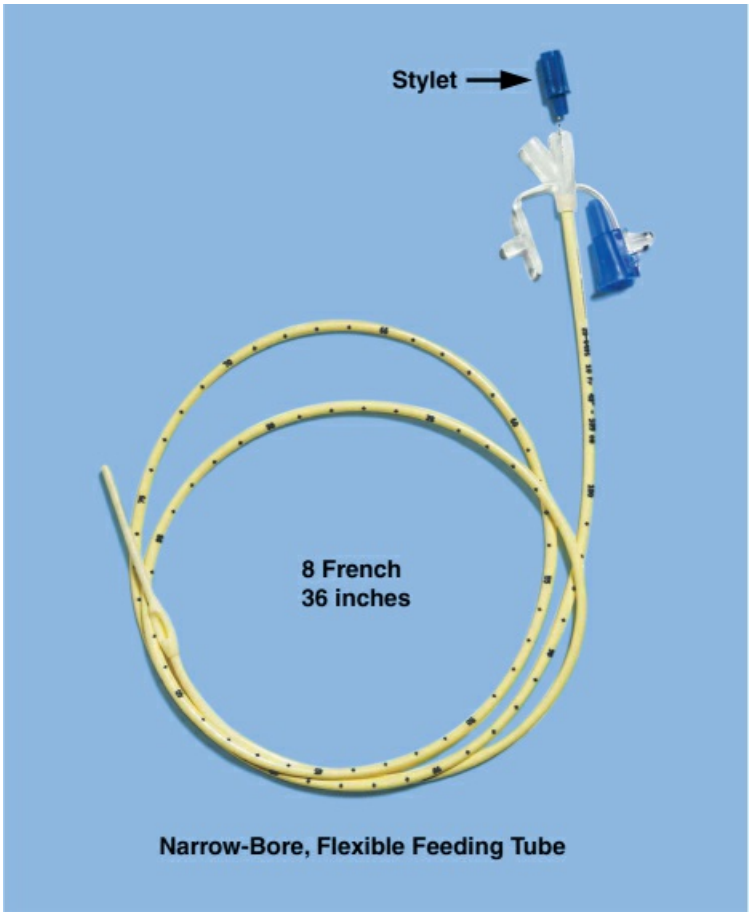


FIGURE 49.1 A narrow-bore, flexible feeding tube with a stylet to facilitate tube placement.

Feeding Tube Placement

Feeding into the stomach is recommended, unless there is evidence of regurgitation and aspiration of feeding formula into the lungs (see later). The distance required to reach the stomach can be estimated by measuring the distance from the tip of the nose to the earlobe and then to the xiphoid process. This “NEX” distance is typically 50–60 cm (29).

The major complication of feeding tube insertion is advancement of the tube into the lungs, which occurs in about 1% of insertions (30). Although a relatively uncommon occurrence, misplaced feeding tubes are a significant source of morbidity and even mortality, especially if there is a delay in detection (31). The problem is that critically ill patients often do not cough or show any signs of distress when a feeding tube enters the lungs (which is especially the case in intubated patients who are heavily sedated), so a portable chest x-ray is required after every insertion to determine if the tube is properly placed. *Feedings should not commence until proper tube placement has been verified*.

Chest Radiography

The appearance of a properly placed feeding tube on a portable chest x-ray is shown in Figure 49.2 (31). The tube runs a straight course down the mediastinum, and bisects the angle created by the tracheal bifurcation. This is evidence that the tube is in the esophagus, and not in the lungs. Placement in the stomach is likely if the tip of the tube is below the dome of the left hemidiaphragm. The optimal tip position is about 10 cm below the dome of the hemidiaphragm (32), which is far enough from the gastroesophageal junction to reduce the risk of reflux into the esophagus.

The Whoosh Test

One of the traditional practices for evaluating tube placement is to push air through the feeding tube and listen for a “whoosh” of air with the stethoscope placed in the epigastric region. This is taken as evidence that the tube is in the stomach, but this is not necessarily the case. In fact, the presence of a whoosh sound does not confirm tube placement in the stomach because a jet of air in the lungs can be heard in the epigastrium (32–34). Despite its popularity, *this test can be misleading, and should be abandoned*.

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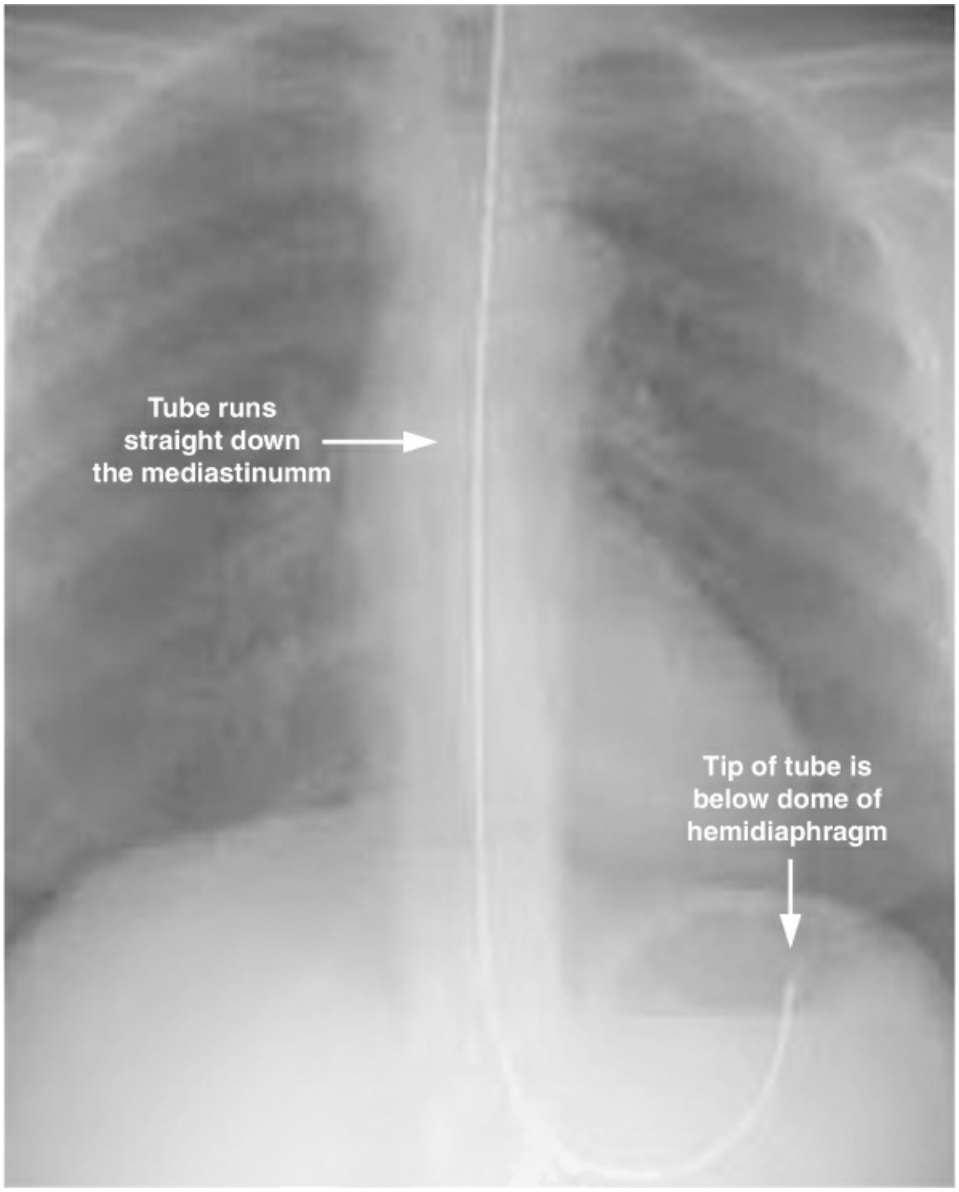


FIGURE 49.2 Portable chest x-ray showing the proper placement of a nasogastric feeding tube. Feedings should not commence until proper placement of the tube is verified radiographically. Image from Reference 31.

Ultrasound

Ultrasound imaging has a sensitivity of 96%, and a specificity of 91%, for confirming the placement of feeding tubes in the stomach (35). However, there is uncertainty about the utility of ultrasound for detecting misplaced feeding tubes, and as a result, ultrasound has not replaced chest radiography for evaluating feeding tube placement.

Misplaced Feeding Tubes

As mentioned earlier, entry of a feeding tube into the airways is usually a covert event, and becomes evident only when the post-procedural chest x-ray looks like the one in Figure 49.3 (36). Note that the tube veers to the right at about the level of the tracheal bifurcation (because it

is following the course of the right mainstem bronchus), in contrast to the straight course of the feeding tube in Figure 49.2. Because they are narrow-bore tubes that are advanced with a rigid stylet, feeding tubes can be advanced far into the lungs, and can break through into the pleural space to produce a pneumothorax (37).

End-Tidal CO_2

Early detection of feeding tubes entering the lungs is possible by monitoring the CO_2 in the infusion port of the feeding tube as it is being advanced. The appearance of a CO_2 waveform, or an increase in PCO_2 to 15 mm Hg, is evidence that the feeding tube has entered the lungs (38). The pooled results of 16 studies has shown that end-tidal CO_2 monitoring has a sensitivity of 96% and a specificity of 99% for the detection of feeding tube misplacement in the lungs (39).

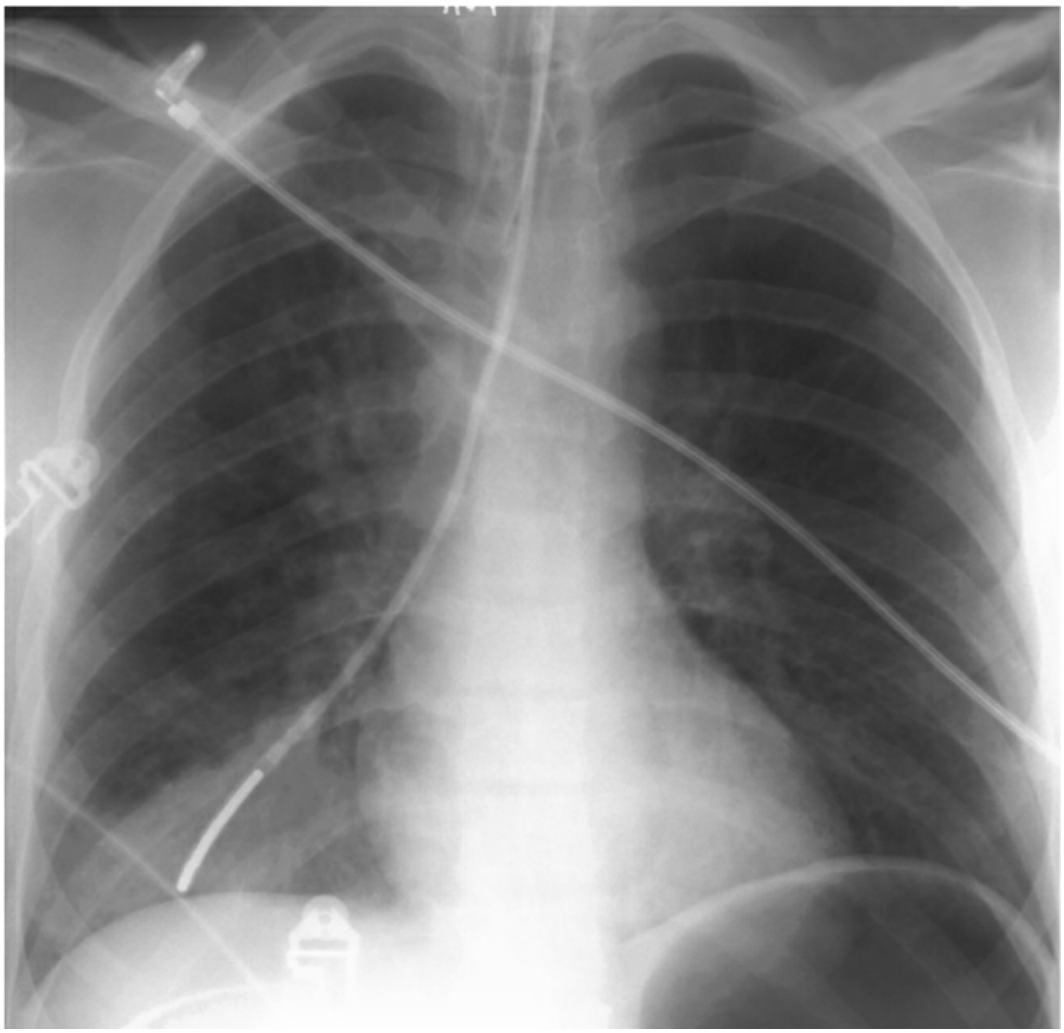


FIGURE 49.3 Portable chest x-ray showing a feeding tube in the right lung. Image from Reference 36, digitally retouched.

End-tidal CO_2 monitoring has one limitation that prevents it from replacing chest radiography: i.e., it does not detect feeding tube misplacement in the esophagus (e.g., from

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coiling). However, the early detection of feeding tube entry into the trachea can help to prevent lung injury, and pneumothoraces.

Gastric vs Duodenal Placement

The consensus recommendation is to infuse the liquid feeding formulas into the stomach rather the proximal duodenum. This recommendation is based on studies showing that *there is no difference in the risk of aspiration with gastric versus duodenal feedings* (40,41), and gastric feedings are much easier to deliver. However, duodenal feeding is a consideration in patients with problematic regurgitation and aspiration of gastric feedings.

COMPLICATIONS OF ENTERAL NUTRITION

The most cited complications of enteral tube feeding include (*a*) occlusion of the feeding tubes, (*b*) regurgitation and aspiration of the feeding formula, (*c*) diarrhea, and (*d*) inadequate nutrition support from frequent interruptions of the tube feeding.

Tube Occlusion

Narrow bore feeding tubes can become occluded by protein precipitates that form when acidic gastric secretions reflux into the feeding tubes (42). Standard preventive measures include flushing the feeding tubes with 30 mL of water every 4 hours, and using a 10 mL water flush after medications are instilled.

Restoring Patency

If flow through the feeding tube is sluggish, flushing the tube with warm water can restore flow in 30% of cases (42). If this is ineffective, *pancreatic enzyme* (Viokase) can be used as follows (43):

Regimen: Dissolve 1 tablet of Viokase and 1 tablet of sodium carbonate (324 mg) in 5 mL of water. Inject this mixture into the feeding tube and clamp for 5 minutes. Follow with a warm water flush. This should relieve the obstruction in about 75% of cases (43).

If the feeding tube is completely occluded, attempt to pass a flexible wire through the tube to clear the obstruction. If this is unsuccessful, replace the feeding tube without delay.

Regurgitation/Aspiration

The most feared complication of tube feedings is regurgitation and aspiration of the liquid feedings into the lungs. Microaspiration occurs in almost half of critically ill patients (44), and there is usually no observable feeding solution in the airways. The heightened risk of regurgitation/aspiration is the result of impaired gastric motility and reduced or absent lower esophageal sphincter tone (45,46). Monitoring gastric residual volumes has been used to determine the risk of regurgitation/aspiration, but with limited success.

Gastric Residual Volume

The traditional practice has been to monitor the gastric residual volume (GRV) routinely (e.g., at least once per shift), and interrupt enteral feeding if the GRV exceeds a threshold volume. (This

volume varies from hospital to hospital, but is usually 150–250 mL.) However, there is no convincing evidence that this practice reduces the risk of aspiration, or otherwise influences outcomes (46), but it does result in frequent interruptions of enteral nutrition, and is a common cause of inadequate nutrition support (2).

RECOMMENDATIONS: The consensus recommendations regarding the GRV are as follows (2,3):

. Routine monitoring of GRV is not necessary, but if it is measured (e.g., when aspiration is suspected), a GRV >500 mL in 6 hours is reason to temporarily interrupt the tube feeding.

. If tube feeding is interrupted, the head of the bed should be elevated to at least 30° above horizontal, and a trial or prokinetic therapy is warranted.

. If the problem persists when tube feedings are reinstituted, the feeding site should be advanced (i.e., from stomach to duodenum, or from duodenum to jejunum).

TABLE 49.5 Prokinetic Therapy

Agent	Dosing Regimens and Comments
Erythromycin	Dosing: 100–200 mg IV every 8 hrs for 3 days. Comment: The most effective prokinetic agent, but tachyphylaxis occurs after a few days. Three-day Rx does not affect the resident microbiota.
Metoclopramide	Dosing: 10 mg IV every 8 hrs for 3 days. Comment: Less effective than erythromycin, and its efficacy wanes after a few days.
Enteral Naloxone	Dosing: 8 mg via nasogastric tube every 6 hrs. Comment: Blocks opiate receptors in the bowel without antagonizing the analgesic effects of opiates. Used only for gastric dysmotility related to opiates.

Prokinetic Therapy

The prokinetic agents and recommended dosing regimens are shown in Table 49.5. Prokinetic therapy can produce short-term improvement in indices of gastric motility, but the clinical significance of these effects has been difficult to demonstrate (47). The following information deserves emphasis.

. *Erythromycin* promotes gastric emptying by stimulating motilin receptors in the GI tract (48). It is the most effective prokinetic agent (49), and *is recommended as first-line therapy for gastric dysmotility* (3). The effectiveness of erythromycin wanes over a few days, so no more than 3 days of therapy is recommended. There is no evidence that 3 days of erythromycin has any influence on the resident microbiota.

. Metoclopramide promotes gastric emptying by antagonizing the actions of dopamine in the GI tract. It is less effective than erythromycin (49), and is more effective when given in combination with erythromycin (50). Its effectiveness wanes after a few days, and only three days of treatment is recommended (3).

. Combined therapy with erythromycin and metoclopramide is more effective than treatment with either drug alone (50). However, both drugs prolong the QT interval, and this must be

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considered when deciding about combination therapy.

. For gastric dysmotility related to opiates, direct intragastric administration of the opioid antagonist *naloxone* can selectively block opioid receptors in the bowel and promote gastric emptying without antagonizing the analgesic effects of the opiate (51). The clinical experience with oral naloxone is limited.

Diarrhea

Diarrhea is reported in as many as 70% of patients receiving enteral tube feedings (52). The liquid feeding formulas can play a role because they are often hyperosmotic to plasma (see Table 49.2), and thus can pull water into the bowel lumen and promote a watery diarrhea. Other sources of diarrhea are unrelated to the feeding formula, and include antibiotics, *Clostridium difficile* infection, and liquid drug preparations.

Liquid Drug Preparations

Liquid preparations are favored for drug delivery through narrow-bore feeding tubes because there is less risk of obstruction. However, liquid preparations have two features that create a risk for diarrhea: i.e., they can be extremely hyperosmolar (≥3,000 mosm/kg H_2O), and they often contain sorbitol (to improve palatability), a well-known laxative that draws water into the bowel lumen (53). Table 49.6 includes a list of diarrhea-prone liquid preparations that might be used in ICU patients.

Management

The management of diarrhea in the ICU can be challenging because of the multiple potential sources. Tube feeding-related management could include the following: (*a*) choose a feeding formula with a lower osmolality, if possible, (*b*) use an elemental formula that has predigested proteins and lipids, which are easier to move into the bloodstream, and (*c*) avoid the liquid drug preparations in Table 49.6, if possible. Although a fermentable fiber might help (by promoting the health of the mucosa), the fiber-containing feeding formulas are a mixture of fermentable and nonfermentable fiber, which provides no advantage in relieving diarrhea.

TABLE 49.6 Liquid Drug Preparations that Promote Diarrhea

> 3,000 mosm/kg H_2O	Contain Sorbitol
Acetaminophen	Acetaminophen
Dexamathasone	Cimetidine
Ferrous sulfate	Isoniazid
Hydroxyzine	Lithium
Metoclopramide	Metoclopramide
Multivitamins	Theophylline
Potassium chloride	Tetracycline
Sodium phosphate	

From Reference 53.

A FINAL WORD

Eating as an Antimicrobial Defense Mechanism

One of the fundamental tenets in critical care is the primacy of the bowel as a source of sepsis, systemic inflammation, and inflammatory organ injury (see Chapter 4). The principal defender in the bowel is the barrier function of the bowel mucosa, which prevents the hordes of enteric pathogens from gaining entry into the body. (Remember that the alimentary canal is outside the body, like the hole in a donut.) And what keeps this mucosal barrier functional is the presence of bulk nutrients in the bowel lumen. The trophic effect of bulk nutrients on the bowel mucosa means that enteral nutrition has a nonnutritive role in protecting against microbial invasion. Stated another way, *eating can be viewed as an antimicrobial defense mechanism*. Interesting concept.

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Chapter 50

Parenteral Nutrition

To lengthen thy life, lessen thy meals.

Benjamin Franklin (*a*)

The intravenous route is used for nutrition support when the enteral route is not feasible, or is unable to provide full nutrition support. This chapter describes the basic features of "parenteral nutrition", and demonstrates how to create a parenteral nutrition regimen for individual patients. The most recent clinical practice guidelines and reviews on the subject are included in the bibliography at the end of the chapter (1–4).

SUBSTRATE SOLUTIONS

Dextrose Solutions

Standard nutrition support regimens use carbohydrates to supply approximately 70% of the daily (nonprotein) calorie requirements. The carbohydrate source for total parenteral nutrition (TPN) is dextrose (glucose), which is available in the solutions shown in Table 50.1 Because the energy yield from dextrose is relatively low (see Table 48.1), the dextrose solutions must be concentrated to provide enough calories to satisfy daily requirements. The standard solution is 50% dextrose (D_{50}), which provides 1,700 kcal per liter. Concentrated dextrose solutions have a very high osmolarity which mandates infusion through large, central veins.

TABLE 50.1 Intravenous Dextrose Solutions

Strength	Concentration (g/L)	Energy Yield* (kcal/L)	Osmolarity (mosmol/L)
5% (D_5)	50	70	253
10% (D_{10})	100	340	505
20% (D_{20})	200	680	1,080
40% (D_{40})	400	1,360	2,220

50% (D_{50})	500	1,700	2,525

*Based on an oxidative energy yield of 3.4 kcal/g for dextrose in its hydrated form.

Amino Acid Solutions

Protein is provided as amino acid solutions that contain mixtures of essential and nonessential amino acids. (*Note:* Humans can synthesize 11 of the 20 amino acids that are used to build proteins. The remaining 9 amino acids, which are called *essential amino acids,* must be provided in the diet. These amino acids include histidine, isoleucine, leucine, lysine, methionine, threonine, tryptophan, and valine.)

Standard amino acid solutions (e.g., Travasol®) are balanced mixtures of 50% essential amino acids and 50% nonessential amino acids. For full nutrition support, a 10% amino acid solution will provides 100 grams of protein per liter, or 16.5 grams of nitrogen per liter. Amino acid solutions can be mixed together with the dextrose solutions and lipid emulsions.

Branched-Chain Amino Acids

Amino acid solutions enriched with branched-chain amino acids (isoleucine, leucine, and valine), such as *Aminosyn II*® are designed for the following situations:

. In progressive liver failure, aromatic amino acids (i.e., phenylalanine, tryptophan, tyrosine) have been implicated in the pathogenesis of hepatic encephalopathy. Branched-chain amino acids competitively block the transport of aromatic amino acids across the blood-brain barrier (5), and can thereby reduce the risk of hepatic encephalopathy (6).

. Branched-chain amino acids are metabolized in skeletal muscle, and parenteral formulas enriched with branched-chain amino acids are more likely to achieve a positive nitrogen balance during periods of metabolic stress like trauma or sepsis (7).

However, despite these theoretical benefits, *nutrition formulas enriched with branched-chain amino acids have not improved outcomes in any of the conditions for which they are designed* (6,7).

Lipid Emulsions

Lipids are provided as emulsions composed of submicron droplets of cholesterol, phospholipids, and triglycerides. The triglycerides are derived primarily from soybean oil, and the features of a popular soybean oil emulsion are shown in Table 50.2 (8,9). As indicated, these emulsions are rich in essential fatty acids (linoleic and linolenic acid), which must be provided in the diet, and which can prevent essential fatty acid deficiency when they provide 3–4% of the daily caloric requirement (10).

Lipids have the highest energy content (9 kcal/g) of the nutritional fuels, and lipid emulsions are used to provide about 30% of the daily energy needs. The soybean emulsions are available in 10% and 20% strengths (the percentage refers to grams of triglyceride per 100 mL of solution): the 10% emulsions provide 1 kcal/mL, and the 20% emulsions provide 2 kcal/mL (8). Unlike the hypertonic dextrose solutions, lipid emulsions are roughly isotonic to plasma, and *can be infused through peripheral veins*. This allows lipids to be used as a major source of calories in protein-sparing nutritional regimens, which are delivered via peripheral veins (as described later in the

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chapter). However, for full parenteral nutrition support, lipid emulsions are added to the glucose and protein solutions to create a 3-in-1 nutrient admixture.

Newer Lipid Emulsions

The polyunsaturated fatty acids in soybean oil are needed to prevent essential fatty acid deficiency, but they have also been implicated in the adverse effects of nutritional lipids, which include hypertriglyceridemia, inflammatory injury, and lipid overload syndrome (which are described later in the chapter). In an attempt to reduce the risk of these complications, lipid emulsions have been introduced that incorporate less damaging lipids. These newer products are listed below (9).

. *Clinolipid*® is 80% olive oil and 20% soybean oil. Olive oil contains medium-chain triglycerides, which are more readily metabolized, and less likely to promote the lipid overload syndrome (11).

. *Smoflipid*® is a mixture of 30% soybean oil, 30% medium-chain triglycerides, 20% olive oil, and 15% fish oil. The fish oil is rich in omega-3 fatty acids, which are less likely to promote inflammatory injury.

. *Omegaven*® is pure fish oil, which has the benefits of omega-3 fatty acids, but will not prevent essential fatty acid deficiency unless combined with lipids from soybean oil.

The experience with the above products is limited at the present time.

TABLE 50.2 Characteristics of Lipid Emulsions Based on Soybean Oil

Product:	Intralipid®
Composition:	1. 50% is linoleic acid, a ω-6 fatty acid and essential fatty acid. 2. 25 % is oleic acid, a ω-9 fatty acid. 3. 10% is α-linolenic acid, a ω-3 fatty acid and essential fatty acid.
Strength:	10% (10 g/100 mL) and 20% (20 g/100 mL)
Energy Yield:	1 kcal/mL for 10% solution 2 kcal/mL for 20% solution
Osmolarity:	260 mosm/L
Max Dosage:	2.5 g/kg/day (or 12.5 mL/kg/day of 20% Intralipid)
Complications:	Hypertriglyceridemia, inflammation, lipid overload syndrome

ADDITIVES

Commercially available mixtures of electrolytes, vitamins, and trace elements are added directly to the 3-in-1 nutrient admixture.

Electrolytes

There are numerous electrolyte mixtures that are commercially available. Most have a volume of 20 mL, and contain sodium, chloride, potassium, and magnesium. You must check the mixture used at your hospital to determine if additional electrolytes must be added. Most hospitals have special order forms where the desired electrolyte changes can be specified.

Vitamins

Aqueous multivitamin preparations will provide the normal daily requirements for most vitamins (see Table 48.3) (16). The daily vitamin requirements in ICU patients are likely to be higher than normal, which explains why vitamin deficiencies are commonplace in ICU patients (see the section on "Vitamin Requirements" in Chapter 48).

Trace Elements

A variety of trace element additives are available, and one of these is shown in Table 50.3, along with the recommended daily requirement for the included trace elements. Note that the daily requirements for 5 of the 7 trace elements are not satisfied by the dosing in this product. The consequences of this mismatch are not known.

TABLE 50.3 Trace Element Preparation and Daily Requirements

Trace Element	Daily Requirement†	Multi-Trace 5 Concentrated§
Chromium	30 µg	10 µg
Copper	900 µg	1 µg
Iodine	150 µg	—
Iron	8 mg	—
Manganese	2.3 mg	0.5 mg/d
Selenium	55 µg	60
Zinc	11 mg	5

†From the National Institutes of Health, Office of Dietary Supplements. (See legend of Table 48.5 for the website of origin.)

§Product description, America Regent, Inc.

CREATING A TPN REGIMEN

The following is a stepwise approach to creating a standard TPN regimen for an individual patient. The patient in this example will be a 70 kg adult who is not malnourished and has no volume restrictions.

Step 1

The initial task is to determine the daily requirement for calories and protein. This is described in

Chapter 48, and is briefly reviewed here. The daily energy requirement, or resting energy expenditure (REE), can be measured by indirect calorimetry, or it can be estimated using a simple formula: REE = 25 × weight (in kg). The daily protein requirements can be estimated as 1.2–2.0 grams per kg body weight. You can use actual body weight in these estimates as long as actual body weight is not greater than 125% of the ideal weight (see Appendix 2 for ideal body weights). If actual body weight exceeds 125% of ideal body weight, you can use the adjusted body weight (which is derived in Equation 48.3).

For the 70 kg patient, we'll use a daily protein requirement of 1.4 g/kg. Therefore, the daily requirement for calories and protein will be:

$$\text{Calories: } 25 \times 70 = 1{,}750 \text{ kcal/day}$$

$$\text{Protein: } 1.4 \times 70 = 98 \text{ grams/day} \qquad (49.1)$$

Note: If propofol is being infused, you should determine the calories provided by propofol and subtract those calories from the daily caloric requirement. Propofol is infused in a 10% lipid emulsion, which has a caloric density equivalent to 10% Intralipid (1 kcal/mL). Therefore, the hourly infusion rate of propofol (mL/hr) is equivalent to the hourly yield of calories (kcal/hr).

Step 2

The next step is to determine the volume of the amino acid solution needed to meet the daily protein requirement. A 10% amino acid solution has 100 grams of protein per liter, so one liter of the amino acid solution will satisfy the estimated daily protein requirement (i.e., 98 grams/day).

Step 3

The next step is to determine how to deliver the daily caloric requirement, which should be provided as nonprotein calories in the dextrose solution and lipid emulsion. Start with the lipid emulsion. The standard lipid emulsion in many hospital pharmacies is a 20% emulsion, which provides 2 kcal/mL. Therefore, 250 mL of a 20% lipid emulsion will provide 500 kcal. That leaves 1,750 – 500 = 1,250 kcal that must be provided by dextrose. One liter of 50% dextrose provides 1,700 kcal, or 1.7 kcal/mL, so the volume of D_{50} needed to provide 1,250 kcal is 1,250/1.7 = 735 mL.

Therefore, the parenteral nutrition formulation for this example would be an admixture of the following:

1. 1 liter of 10% amino acid solution
2. 250 mL of a 20% lipid emulsion
3. 735 mL of a 50% dextrose solution (D_{50})
4. 30 mL for vitamins, electrolytes, and trace elements

Step 4

The final step is determining the infusion rate of the nutrient admixture. The total volume of the admixture is 2,015 mL, which is infused over 24 hours, so the hourly infusion rate is 2,015/24 = 84 mL/hr. However, in patients who are at risk for the refeeding syndrome (see later), the American Society of Parenteral and Enteral Nutrition recommends starting at 40–50% of the

nutritional goal, and advancing to full nutrition support as allowed by the serum levels of phosphate, potassium, and magnesium (usually 4–7 days) (12). For this example, starting at 50% of the nutritional goal corresponds to an infusion rate of 42 mL/hr.

COMPLICATIONS

The following complications do not include those associated with central venous catheters, which are required for full parenteral nutrition. Catheter-related complications are described in Chapters 2 and 3.

Carbohydrate Complications

Hyperglycemia

Hyperglycemia is common during total parenteral nutrition (TPN): e.g., in one report, blood glucose levels above 200 mg/dL were recorded in 45% of the patients receiving TPN (13). Tight glycemic control is not recommended in critically ill patients because of the risk of hypoglycemia, which has more serious consequences than hyperglycemia (14). The current recommendation for hospitalized patients is *a target range of 140–180 mg/dL for blood glucose* (15).

INSULIN: Insulin is often necessary for glycemic control during TPN, and Table 50.4 shows the variety of biosynthetic insulins that are available (16). A continuous infusion of regular insulin is preferred for critically ill patients (17), and this can often be accomplished by adding regular insulin to the feeding admixture, and using subcutaneous insulin (lispro) as needed for glycemic control. When patients are clinically stable, half of the daily insulin requirement can be converted to long-acting glargine insulin, which is given once daily.

TABLE 50.4 Insulin Preparations

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Type	Name	Onset*	Peak	Duration
Rapid-Acting	Aspart	10–20 min	1–3 hr	3–5 hr
Rapid-Acting	Glulisine	25 min	45–50 min	4–5 hr
Rapid-Acting	Lispro	15–30 min	0.5–2.5 hr	3–6 hr
Short-Acting	Regular	30–60 min	1–5 hr	6–10 hr
Intermediate	NPH	1–2 hr	6–14 hr	16–24 hr
Long-Acting	Glargine	1 hr	2–20 hr	24 hr

*Onset after subcutaneous injection. From Reference 17.

Hypophosphatemia

The movement of glucose into cells is associated with a similar movement of phosphate into cells, which provides phosphate for cofactors (e.g., thiamine pyrophosphate) that participate in glucose metabolism. This intracellular shift of phosphate can result in progressive hypophosphatemia, as shown in Figure 38.2 (18). *Glucose-induced hypophosphatemia is a major component of the refeeding syndrome* (see later), and this mandates attention to monitoring serum phosphate levels when TPN is initiated.

Hypokalemia

Glucose movement into cells is also accompanied by an intracellular shift of K^+ (which is the basis for the use of glucose and insulin to treat severe hyperkalemia). This effect is usually transient, but continued glucose loading during TPN can lead to persistent hypokalemia. Glucose-induced hypokalemia is another component of the refeeding syndrome (see later).

Hypercapnia

Overfeeding promotes CO_2 retention in patients with respiratory insufficiency (19). This has been attributed to glucose metabolism, which produces more CO_2 than protein or lipid metabolism (i.e., glucose has a higher RQ than proteins or lipids, as shown in Table 48.1). However, CO_2 retention appears to be a consequence of overfeeding, and not just overfeeding with carbohydrates (19).

Lipid Complications

Lipid emulsions have the potential for several deleterious effects, including hypertriglyceridemia, inflammatory injury, hepatic steatosis, and the lipid overload syndrome.

Inflammatory Injury

The soybean oil used in standard lipid emulsions is rich in omega-6 and omega-9 polyunsaturated fatty acids that serve as precursors to the formation of eicosanoids (prostaglandins and leukotrienes), which have a pivotal role in the inflammatory response (4,9). The inflammatory potential of soybean–based lipid emulsions is demonstrated by one of the animal models for the acute respiratory distress syndrome (an inflammatory lung injury) which involves the infusion of oleic acid (an omega-9 fatty acid in soybean oil) into the pulmonary arteries (20). This would explain why lipid infusions in TPN formulations are associated with impaired oxygenation, and can prolong the respiratory failure (21,22). The omega-3 fatty acids in fish oil have anti-inflammatory activity, and some of the newer lipid formulations mentioned earlier (e.g., Smoflipid®) have incorporated fish oils to take advantage of this activity.

Hepatic Steatosis

Fat accumulation in the liver (hepatic steatosis) can occur, but is more common in patients receiving long-term TPN. Contributing factors include the ratio of omega-6 to omega-3 fatty acids in the lipid emulsion, and the content of phytosterols (plant sterols) in lipid emulsions (23). The hepatic steatosis can progress to a chronic hepatitis and even cirrhosis if the TPN is very prolonged (24).

Cholestasis

TPN is also associated with cholestasis, which can lead to acalculous cholecystitis (24). This is attributed to the absence of lipids in the proximal small bowel, which prevents cholecystokinin-mediated contraction of the gallbladder.

Lipid Overload Syndrome

The lipid overload syndrome is an uncommon condition that occurs when the lipid-induced increase in serum triglycerides overwhelms the ability to break down the triglycerides, resulting in a clinical syndrome that can include fever, headache, irritability, respiratory distress, jaundice, pancreatitis, pancytopenia, and coagulopathy (25,26). This is reported more frequently in children, and it can resolve when the lipid infusions are temporarily discontinued (26).

Refeeding Syndrome

The refeeding syndrome was first described in World War II, when aggressive feeding of malnourished prisoners of war and concentration camp victims resulted in considerable morbidity and even mortality.

What is it?

The refeeding syndrome is characterized by the following abnormalities (see Table 50.5): hypophosphatemia, hypokalemia, hypomagnesemia, thiamine deficiency, and renal sodium retention (12). Many of the electrolyte abnormalities, along with the thiamine deficiency, are attributed to the introduction of a glucose load in patients with borderline or depleted stores of potassium, phosphate, magnesium, and thiamine. The sodium retention could be the result of hypoalbuminemia from malnutrition.

The clinical manifestations are a consequence of the electrolyte and vitamin deficiencies, and include generalized weakness (from hypophosphatemia), mental status changes and lactic

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acidosis (from thiamine deficiency), tachyarrhythmias (from hypokalemia and hypomagnesemia), and pulmonary edema (from sodium retention) (12).

TABLE 50.5 The Refeeding Syndrome

Mechanisms	Possible Manifestations
Hypophosphatemia	Generalized Weakness
Hypokalemia	Hypotension
Hypomagnesemia	Altered Mentation
Hydroxyzine	Lactic Acidosis
Sodium Retention	Tachyarrhythmias
Multivitamins	Pulmonary Edema
Thiamine Deficiency	

From Reference 12.

Predisposing Conditions

The following conditions have been identified as high-risk for the refeeding syndrome (12): weight loss of 5% in one month, or >10% in 6 months, body mass index (BMI) ≤18.5 kg/m^2, poor nutrient intake for the past 5 days, low baseline levels of serum potassium, phosphate, or magnesium, or any patient with a high risk of malnutrition (e.g., from drug abuse).

Recommendations

The following recommendations are for patients who are high-risk for the refeeding syndrome (12):

. Before nutrition support is started, electrolyte abnormalities should be corrected, and thiamine should be started. The American Society of Parenteral and Enteral Nutrition recommends 100–200 mg thiamine daily (12), but this is inadequate for correcting thiamine deficiency (27), and the replacement protocol in Table 48.4 is more appropriate.

. Nutrition support should be started at 40–50% of the estimated nutritional needs (or 10–15 kcal/kg per day), or glucose infusions can be started at 150 grams daily (which is 300 mL of D_{50}). The nutrition regimen is then slowly advanced over the next 4–7 days, while closely monitoring the serum levels of phosphate, potassium, and magnesium.

PERIPHERAL PARENTERAL NUTRITION

Peripheral parenteral nutrition (PPN) is a truncated form of TPN that can be used to provide nonprotein calories in amounts that will spare the breakdown of proteins to provide energy (i.e., *protein-sparing* nutrition support). PPN can be used as a supplement to enteral feeding, or as a source of calories during brief periods of inadequate nutrition. It is not intended for hypercatabolic or malnourished patients, who need full nutritional support,

The osmolarity of peripheral vein infusates should be kept below 900 mosm/L, with a pH between 7.2 and 7.4, to minimize the risk of vascular damage (28,29). This requires dilute amino acid and dextrose solutions, which limits nutrient intake. However, lipid emulsions are isotonic

to plasma, and lipids can be used to provide a considerable portion of the nonprotein calories in PPN.

Method

A popular regimen for PPN is a mixture of 3% amino acids and 20% dextrose (final concentration of 1.5% amino acids and 10% dextrose), which has an osmolarity of 500 mosm/L. The dextrose will provide 340 kcal/L, so 2.5 L of the mixture will provide 850 kcal. If 250 mL of 20% Intralipid is added to the regimen (adding 500 kcal), the total nonprotein calories will increase to 1,350 kcal/day. This should be close to the nonprotein calorie requirement of an average size, unstressed adult (20 kcal/kg/day).

A FINAL WORD

The final word on parenteral nutrition is . . . *AVOID* . . . whenever possible. For an explanation, see the first section of Chapter 49.

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Chapter 51

Adrenal and Thyroid Dysfunction

Judge a tree from its fruit, not from the leaves.

Euripides

(484–406 BC)

Adrenal and thyroid disorders are rarely the primary reason for admission to an ICU. However, critical illness can influence adrenal and thyroid function, and there is concern that this influence can have a negative impact on outcomes in critically ill patients. This chapter describes the spectrum of adrenal and thyroid disorders that occur in critically ill patients, and how critical illness influences adrenal and thyroid function.

ADRENAL SUPPRESSION IN THE ICU

The adrenal gland plays a major role in the adaptive response to stress. The adrenal cortex releases glucocorticoids and mineralocorticoids that promote glucose availability and maintain extracellular volume, while the adrenal medulla releases catecholamines that support the circulation. Attenuation or loss of this adrenal response leads to hemodynamic instability, volume depletion, and defective energy metabolism (1,2). Adrenal insufficiency can remain silent until the adrenal gland is called on to respond to a physiologic stress. When this occurs, adrenal insufficiency becomes an occult catalyst that speeds the progression of acute, life threatening conditions.

Adrenal Activity

The activity of the adrenal glands is governed by the release of adrenocorticotrophic hormone (ACTH) from the anterior pituitary gland, which, in turn, is governed by the production of corticotrophin-releasing hormone (CRH) in the hypothalamus (see Figure 51.1). Adrenal insufficiency can be the result of suppression at the hypothalamic-pituitary level, or primary suppression of adrenal gland activity.

Cortisol

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Cortisol (hydrocortisone) is the major glucocorticoid released by the adrenal cortex. The daily production of cortisol in the normal (unstressed) adult is 15–25 mg/day, and this can increase up to 350 mg/day during periods of maximum physiological stress (3).

PLASMA CORTISOL: About 90% of the cortisol in plasma is bound to corticosteroid-binding globulin (CBG) and albumin, while the remaining 10% is the free or biologically active form (3). The commercial assay for plasma cortisol measures both bound and unbound fractions; i.e., total cortisol. This assay can be misleading in acutely ill patients because plasma levels of CBG fall by as much as 50% during an acute illness (3). Studies of the correlation between free and total cortisol in critically ill patients have shown inconsistent results (1). However, free cortisol levels are not readily available, so total cortisol levels are recommended for assessing adrenal function (1).

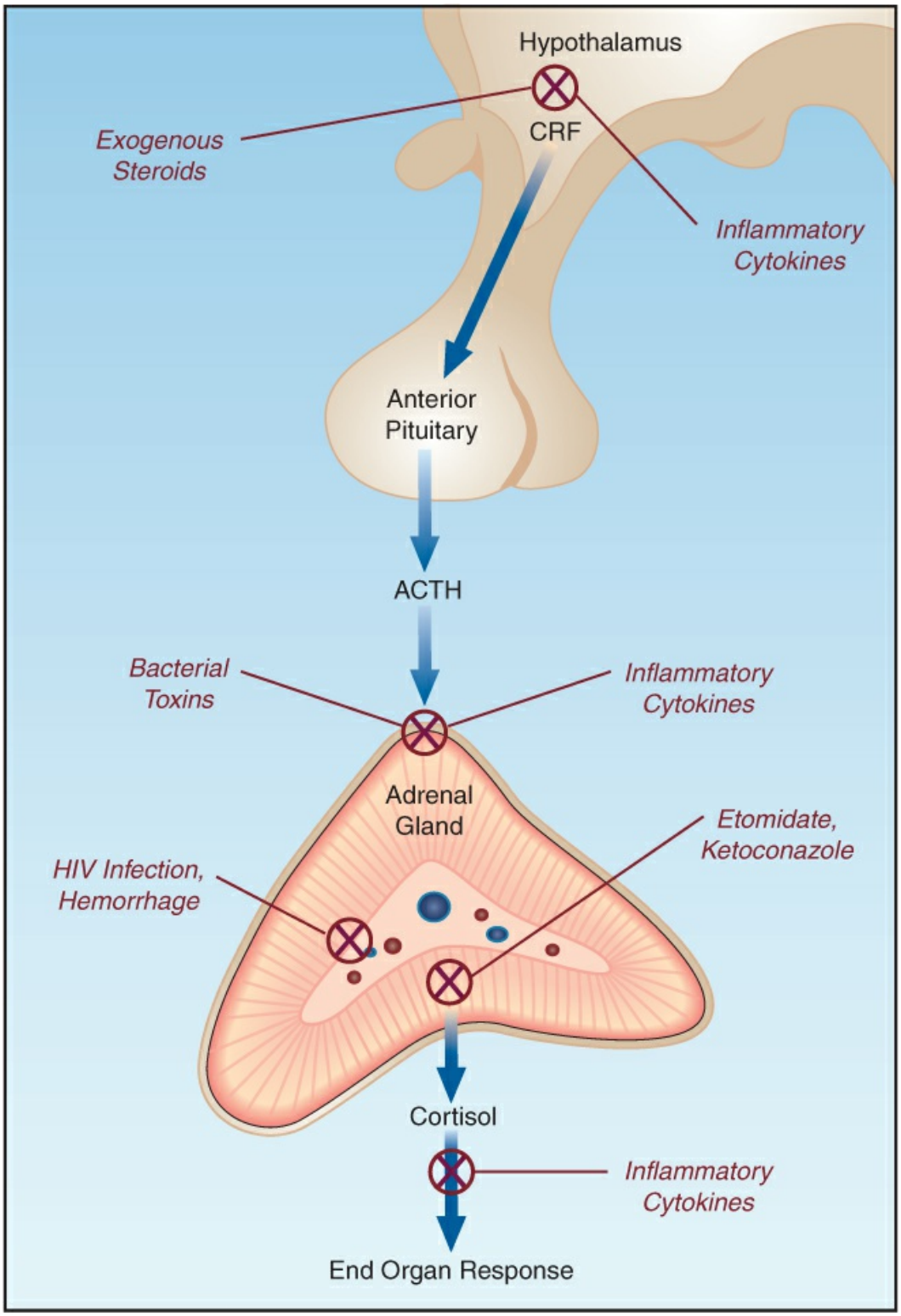


FIGURE 51.1 Mechanisms of adrenal suppression in ICU patients. CRF = corticotrophin-releasing hormone, ACTH = adrenocorticotrophic hormone.

Critically Ill Patients

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Adrenal insufficiency is common in critically ill patients. The overall incidence is 10–20% (4), but rates as high as 60% have been reported in patients with severe sepsis and septic shock (5). The adrenal suppression in critically ill patients is *often reversible*, and is called *critical illness-related corticosteroid insufficiency* (CIRCI) (1,3). The mechanisms involved in CIRCI are complex, and not fully elucidated; Figure 51.1 shows some of the known mechanisms (3–6). As indicated, the systemic inflammatory response plays a major role in CIRCI. *Suppression at the hypothalamic-pituitary level* is particularly prominent, and *is responsible for as many as 75% of the cases of adrenal suppression in patients with severe sepsis and septic shock* (5).

Predisposing Conditions

Severe sepsis and septic shock are the leading causes of adrenal suppression in critically ill patients. Specific infectious causes of adrenal suppression include HIV infection, systemic fungal infections, and meningococcemia (which can result in adrenal hemorrhage) (3,6).

Noninfectious sources of adrenal suppression in ICU patients include: (*a*) abrupt discontinuation of chronic steroid therapy, (*b*) adrenal hemorrhage from disseminated intravascular coagulation (DIC) or anticoagulant therapy, and (*c*) drugs that inhibit the synthesis of cortisol (e.g., etomidate and ketoconazole) or accelerate the metabolism of cortisol (e.g., phenytoin or rifampin) (3,6).

Clinical Manifestations

The principal manifestation of adrenal suppression in critically ill patients is *hypotension that is refractory to volume resuscitation* (3–5). The typical electrolyte abnormalities that accompany chronic adrenal insufficiency (i.e., hyponatremia and hyperkalemia), are uncommon in the adrenal suppression associated with critical illness.

Diagnosis

Adrenal suppression should be suspected in any ICU patient with labile blood pressure, or with hypotension of unclear etiology that does not respond to volume resuscitation.

Rapid ACTH Stimulation Test

A popular (but often unnecessary) test of adrenal function in ICU patients is the rapid ACTH stimulation test. This test can be performed at any time of the day or night. A blood sample is obtained for a baseline (random) plasma cortisol level, and the patient is given synthetic ACTH (Cosyntropin) intravenously in a dose of 250 µg. Sixty minutes after the ACTH injection, a second blood sample is obtained for a repeat plasma cortisol level. The interpretation of the test results is described next.

Diagnostic Criteria

Unfortunately, the diagnosis of adrenal suppression in critically ill patients is steeped in uncertainty. The following recommendations represent the current consensus opinion on the subject (1).

. The best predictor of adrenal suppression in critically ill patients is a random plasma cortisol level <10 µg/dL, *or* an increment in plasma cortisol of <9 µg/dL after the intravenous injection

of synthetic ACTH (250 µg).

. In patients with septic shock, plasma cortisol levels are not necessary for identifying patients who might benefit from corticosteroid therapy. In these patients, a trial of intravenous hydrocortisone is recommended when hypotension is refractory to volume resuscitation and moderate doses of a vasopressor agent (1).

The popular approach is to forego the rapid ACTH stimulation test and rely on the random plasma cortisol level for the diagnosis of adrenal suppression (except in patients with septic shock). A random plasma cortisol level that is ≥35 µg/dL is evidence of normal or adequate adrenal function, while a baseline cortisol level that is below 10 µg/dL is evidence of adrenal suppression. A rapid ACTH stimulation test can be performed when the random serum cortisol level is indeterminate. However, *a normal response to ACTH (i.e., an increment in serum cortisol of ≥9 µg/dL) does not eliminate the possibility of secondary adrenal suppression from hypothalamic-pituitary dysfunction* (which may be more common than suspected in ICU patients). A plasma ACTH level is needed for the diagnosis of secondary adrenal insufficiency (2), but this is rarely performed in the ICU setting (for unclear reasons).

Treatment

The treatment of critical illness-related adrenal suppression is *intravenous hydrocortisone in a dose of 200–300 mg daily* (i.e., 100 mg every 8 hours) (4). The addition of a mineralocorticoid (i.e., fludrocortisone, 50 µg orally once daily) is considered optional (4), because hydrocortisone has excellent mineralocorticoid activity (see Table 51.1) (7).

TABLE 51.1 Corticosteroid Comparisons

Corticosteroid	Equivalent Doses	Mineralocorticoid Activity†	Anti-Inflammatory Activity†
Hydrocortisone	20 mg	1	4
Prednisone	5 mg	2	3
Methylprednisolone	4 mg	3	2
Dexamethasone	0.75 mg	4	1

†1 = best, 4 = worst. From Reference 7.

Hydrocortisone can be discontinued after satisfactory resolution of the underlying condition. In septic shock, hydrocortisone can be discontinued when vasopressor therapy is no longer necessary, and serum lactate levels have normalized. A gradual taper of the hydrocortisone dose (over at least a few days) is recommended to prevent a rebound increase in proinflammatory mediators (4).

Steroids as Anti-Inflammatory Agents

Regardless of the presence or absence of CIRCI, corticosteroids are used liberally as anti-inflammatory agents in conditions like septic shock, acute respiratory distress syndrome, autoimmune diseases, and severe infections like pneumonias and meningitis. However, the popular steroid for these conditions is methylprednisolone, despite the fact that *dexamethasone is*

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the most potent anti-inflammatory corticosteroid (see Table 51.1). The reason for this discrepancy is unclear.

EVALUATION OF THYROID FUNCTION

Laboratory tests of thyroid function can be abnormal in up to 90% of critically ill patients (8). In most cases, the abnormality is a consequence of non-thyroidal (systemic) illness, and is not a sign of pathologic thyroid disease (8,9). This section describes the laboratory evaluation of thyroid function, and explains how to identify non-thyroidal illness as a cause of abnormal thyroid function tests.

Thyroxine (T_4) and Triiodothyronine (T_3)

Thyroxine (T_4) is the principal hormone secreted by the thyroid gland, but the active form is triiodothyronine (T_3), which is formed by deiodination of thyroxine in extrathyroidal tissues. Both T_3 and T_4 are extensively (>99%) bound to plasma proteins (especially thyroxine-binding globulin), and less than 1% of either hormone is present in the free, or biologically active form (10). Because of the potential for alterations in plasma proteins and protein binding in acute illness, free T_4 and T_3 levels are more reliable for assessing thyroid function in ICU patients. Free T_3 levels are not routinely available, so *free T_4 levels are used to evaluate thyroid function in acutely ill patients*.

Thyroid-Stimulating Hormone (TSH)

The plasma level of thyroid-stimulating hormone (TSH) is considered the most reliable test for the evaluation of thyroid illness, as it can distinguish primary from secondary thyroid disorders, and can also identify nonthyroidal influences on thyroid function. The following general rules apply to the interpretation of thyroid function tests:

. In primary thyroidal illness, the plasma TSH will change in the opposite direction to changes in thyroid hormone levels, and in secondary thyroidal illness (due to aberrations in the hypothalamic-pituitary axis), plasma TSH levels will change in the same direction as changes in thyroid hormone levels.

. For nonthyroidal influences on thyroid function tests, TSH levels are usually in the normal range.

Patterns of Abnormal Thyroid Function Tests

Table 51.2 shows the variety of patterns of free T_4 and TSH levels that can occur with abnormal thyroid function. These patterns will be explained in the following sections on hyperthyroidism and hypothyroidism.

TABLE 51.2 Patterns of Abnormal Thyroid Function Tests

Condition	Free T_4	TSH

Normal Range	0.8–1.8 ng/dL	0.3–4.5 mU/mL
Primary Hyperthyroidism	↑	↓
T_3-Toxicosis	NL	↓
Euthyroid Hyperthyroxinemia	↑	NL
Primary Hypothyroidism	↓	↑
Secondary Hypothyroidism	↓	↓
Euthyroid Sick	↓	NL

NL = normal.

THYROTOXICOSIS

Thyrotoxicosis is almost always the result of primary hyperthyroidism, and the most common cause is autoimmune thyroiditis (Graves disease) (11,12).

Clinical Manifestations

The principal manifestations of thyrotoxicosis include agitation, heat intolerance, tachycardia (including atrial fibrillation), and fine tremors. *Elderly patients with hyperthyroidism can be lethargic* rather than agitated; this condition is called *apathetic thyrotoxicosis*. The combination of lethargy and atrial fibrillation is a frequently cited presentation for apathetic thyrotoxicosis in the elderly (13).

Thyroid Storm

An uncommon but severe form of hyperthyroidism known as *thyroid storm* can be precipitated by acute illness or surgery. This condition is characterized by hyperpyrexia (body temperatures can exceed 104° F), severe agitation or delirium, and severe tachycardia with high output heart failure. Advanced cases are associated with obtundation or coma, generalized seizures, and hemodynamic instability. The mortality rate in this condition is as high as 25%, even with treatment (11).

Diagnosis

The plasma TSH assay is the most sensitive and specific diagnostic test for hyperthyroidism, and is recommended as the initial screening test for suspected hyperthyroidism (11). TSH levels are almost undetectable (<0.01 mU/dL) in most cases of primary hyperthyroidism, and a normal TSH level excludes the diagnosis of hyperthyroidism (11). The free T_4 level is typically elevated in cases of overt hyperthyroidism.

T_3-Toxicosis

In mild or subclinical hyperthyroidism, the plasma TSH level is low, but the free T_4 level can be normal. This condition is sometimes called *T_3-toxicosis*, even though T_3 levels can also be normal (11).

Euthyroid Hyperthyroxinemia

Drugs that block the conversion of T_4 to T_3, such as amiodarone (14) and amphetamines (15) can produce a condition called *euthyroid hyperthyroxinemia* where free T_4 levels are elevated, but TSH levels are normal (11). Heparin therapy can also produce this condition by activating lipoprotein lipase, which increases free fatty acids that will displace T_4 from its binding proteins (11).

Management

The drugs used to treat thyrotoxicosis and thyroid storm are presented in Tables 51.3 and 51.4.

β-Receptor Antagonists

Treatment with β-receptor antagonists relieves the tachycardia, agitation, and fine tremors in thyrotoxicosis. *Propranolol* has been the most widely used β-receptor antagonist in hyperthyroidism (see Table 51.3 for the dosing recommendations), and is preferred in thyroid storm because it blocks the conversion of T_4 to T_3 in high doses (11). However, it is a non-selective β-blocker, which makes it less than ideal for patients with asthma or systolic heart failure. More selective β-blockers like *metoprolol* (25–50 mg PO twice or three times daily) can be used for patients with heart failure or a history of asthma. (*Note:* All β-receptor antagonists are contraindicated during acute exacerbations of asthma). Finally, the ultra rapid-acting agent *esmolol* can be given by continuous infusion for rapid control of heart rate.

TABLE 51.3 Beta Blocker Therapy for Thyrotoxicosis and Thyroid Storm

Drug	Dosing Regimens and Comments
Propranolol	Dosing: 10–40 mg PO 3–4 times daily for thyrotoxicosis, and 60–80 mg IV or PO every 4 hrs for thyroid storm. Comment: The preferred β-blocker, because it blocks conversion of T_4 to T_3 in high doses. However, it is a nonselective β-blocker.
Metoprolol	Dosing: 25–50 mg PO 2–3 times daily. Comment: Relative β-1 selectivity, which may be advantageous in patients with heart failure or asthma.†
Esmolol	Dosing: 50–100 μg/kg/min IV by continuous infusion. Comment: Ultra-short acting agent that allows rapid titration of dosage.

†All β-blockers are contraindicated during acute exacerbations of asthma.

Dosing regimens from Reference 11.

Antithyroid Drugs

Two drugs are used to suppress thyroxine production: methimazole and propyl-thiouracil (PTU). Both are given orally. *Methimazole is preferred for the treatment of thyrotoxicosis, while PTU is favored for the treatment of thyroid storm* (11). Uncommon but serious side effects include cholestatic jaundice for methimazole, and fulminant hepatic necrosis plus agranulocytosis for PTU (11). (See Table 51.4 for the dosing regimens for each drug.)

Inorganic Iodine

In severe cases of hyperthyroidism, iodine (which blocks the synthesis and release of T_4) can be added to antithyroid drug therapy. The iodine is given orally as a saturated potassium iodide solution (Lugol's solution). In patients with an iodine allergy, lithium (300 mg orally every 8 hours) can be used as a substitute (16).

Additional Measures in Thyroid Storm

In addition to the above measures, the management of thyroid storm often requires the following measures:

. Aggressive volume resuscitation is often needed to replace fluid losses from vomiting, diarrhea, and heightened insensible fluid loss.

. Thyroid storm can accelerate glucocorticoid metabolism and create a relative adrenal insufficiency. Prophylactic therapy with intravenous hydrocortisone (300 mg IV as a loading dose, followed by 100 mg IV every 8 hours) is recommended (1).

Successful management of thyroid storm also requires treatment of the precipitating event.

TABLE 51.4 Antithyroid Drugs for Thyrotoxicosis and Thyroid Storm

Drug	Dosing Regimens and Comments
Methimazole	Dosing: 10–20 mg PO once daily for thyrotoxicosis, and 60–80 mg PO once daily for thyroid storm. Comment: Blocks synthesis of T_4. Preferred to propylthiouracil for thyrotoxicosis, but not for thyroid storm.
Propylthiouracil	Dosing: 50–150 mg PO TID for thyrotoxicosis, or 500–1,000 mg PO loading dose, then 250 mg PO every 4 hours for thyroid storm. Comment: Blocks both T_4 synthesis and the conversion of T_4 to T_3. Preferred to methimazole for thyroid storm.
Iodine	Dosing: 5 drops of saturated potassium iodide (Lugol's) solution (250 mg iodine) PO every 6 hrs, for severe thyrotoxicosis or thyroid storm. Comment: Blocks synthesis and secretion of T_4. Used in combination with antithyroid drugs.
Hydrocortisone	Dosing: 300 mg IV as a loading dose, then 100 mg IV every 8 hrs. For thyroid storm only. Comment: Prophylaxis for the relative adrenal insufficiency in thyroid storm. Also blocks conversion of T_4 to T_3 in high doses.

Dosing regimens from Reference 11.

HYPOTHYROIDISM

Symptomatic hypothyroidism is reported in 3–7% of the general population in the United States (17). Most cases are the result of chronic autoimmune thyroiditis (Hashimoto's thyroiditis), while less common causes include radioiodine or surgical treatment of hyperthyroidism, hypothalamic-pituitary dysfunction from tumors and hemorrhagic necrosis (Sheehan's

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syndrome), and drugs (lithium, amiodarone, tetracyclines, and antineoplastic agents).

Clinical Manifestations

The clinical manifestations of hypothyroidism are often subtle, and include dry skin, fatigue, muscle cramps, and constipation. Contrary to popular perception, obesity is not a consequence of hypothyroidism (18). More advanced cases can be accompanied by hyponatremia and a skeletal muscle myopathy, with elevations in creatine phosphokinase (CPK), and an increase in the serum creatinine (from creatine released by skeletal muscle) in the absence of renal dysfunction (19).

Cardiovascular

Hypothyroidism can be a source of diastolic heart failure (from impaired relaxation during diastole) and exudative pericardial effusions (from increased capillary permeability) (20). Hypothyroidism also prolongs the Q-T interval, and increases the risk of torsade de pointes (see Figure 19.9) (20).

Myxedema Coma

The most life-threatening presentation of hypothyroidism is a condition characterized by depressed consciousness, hypothermia, hyponatremia, and cardiovascular instability. This condition is referred to as *myxedema coma*, but this is a misnomer, since neither coma nor myxedema is a feature of this condition (21). (Myxedema is nonpitting edema that is associated with hypothyroidism, and is caused by the intradermal accumulation of glycosaminoglycans.)

Diagnosis

Free T_4 levels are always reduced in hypothyroidism (T_3 levels can be normal), while TSH levels are increased (often above 10 mU/dL) in primary hypothyroidism, and are depressed in hypothyroidism due to hypothalamic-pituitary dysfunction (17,18).

Euthyroid Sick

Acute, non-thyroidal illness is associated with low plasma levels of free T_3, which is the result of impaired conversion of T_4 to T_3 in non-thyroidal tissue (8). With increasing severity of illness, both free T_3 and free T_4 levels are depressed, which is the pattern reported in 30–50% of ICU patients (9,10). This condition has also been called the *euthyroid sick syndrome* (8), and is usually associated with normal TSH levels (see Table 51.2).

Treatment

The treatment for mild to moderate hypothyroidism is *levothyroxine* (T_4), which is started at an oral dose of 1.5–1.8 µg/kg/day (22). Therapy is then advanced while monitoring the plasma TSH level. The optimal dose of levothyroxine is the lowest dose that returns the TSH to within the normal range. The usual dose is 75 to 150 µg/day.

Myxedema Coma

Prompt treatment of myxedema coma is mandatory, although the mortality rate is as high as 60%

(22), even with treatment. The recommended therapy is as follows (23):

. Myxedema coma is often accompanied by adrenal insufficiency, so *hydrocortisone* (100 mg IV every 8 hours) is recommended at the outset, with the first dose given before thyroid replacement begins.

. *Levothyroxine* is started with a slow intravenous bolus of 200 to 400 µg, followed by a maintenance dose of 50 to 100 µg orally per day (or 1.6 µg/kg/day). For IV dosing, use 75% of the oral dose.

. Since the conversion of T_4 to T_3 can be depressed in critically ill patients, therapy with *liothyronine* (T_3) can be added, using an IV loading dose of 5–20 µg, followed by a maintenance dose of 2.5–10 µg IV every 8 hours, which is continued until the patient awakens and stabilizes.

Euthyroid Sick

No treatment is recommended for patients with the euthyroid sick syndrome (23), as these patients are not considered to be clinically hypothyroid.

A FINAL WORD

Much Ado About Not Much

The introduction to this chapter included the concern that critical illness has a deleterious influence on adrenal and thyroid function, and that this influence has a negative impact on clinical outcomes in critically ill patients. The following observations do not support this concern.

. Critical illness-related corticosteroid insufficiency (CIRCI) is common in illnesses like septic shock, but there is no evidence that steroid therapy in these illnesses improves outcomes.

. Critical illness can influence plasma thyroid hormone levels (e.g., euthyroid sick syndrome), but this condition is not considered to be clinically relevant, and does not warrant treatment.

Based on these observations, it seems that the interaction between critical illness and adrenal or thyroid function has little relevance or clinical import. (A tree with little fruit, as Euripides might say.)

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